

SIMULASI PROGRAM PLC BERBASIS GX WORKS 2 SEBAGAI SIMULATOR PADA SISTEM AUTOMATION TRASHRACK CLEANER

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NRP : 11-2022-019

PROGRAM STUDI TEKNIK ELEKTRO

2025

Pembimbing I (Itenas)

Ir. Decy Nataliana, M.T.

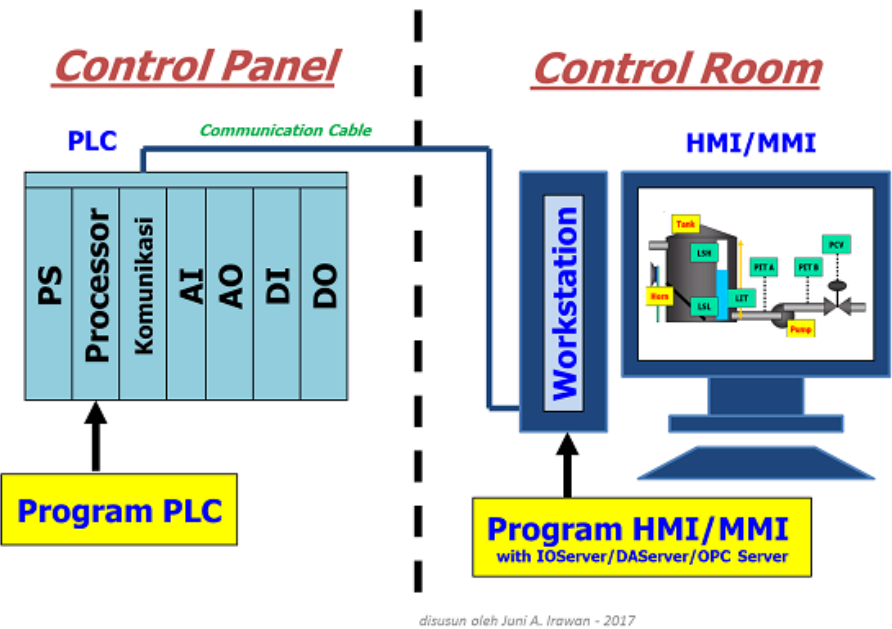
Pembimbing II (PT. Renerconsys)

Mohamad Aji Subekti

Latar Belakang

Perkembangan teknologi otomasi industri telah membawa kemajuan besar dalam pengelolaan infrastruktur, salah satunya pada sistem trashrack cleaner yang digunakan di saluran air. Trashrack cleaner berfungsi untuk membersihkan saringan dari sampah seperti daun, ranting, dan plastik, guna memastikan aliran air tetap optimal.

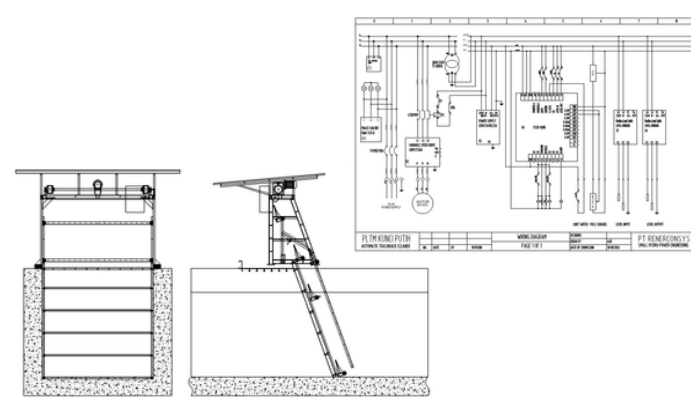
Dalam sistem otomasi modern, trashrack cleaner dikendalikan oleh PLC sebagai pusat logika kontrol otomatis dan HMI sebagai antarmuka visual antara operator dan mesin.



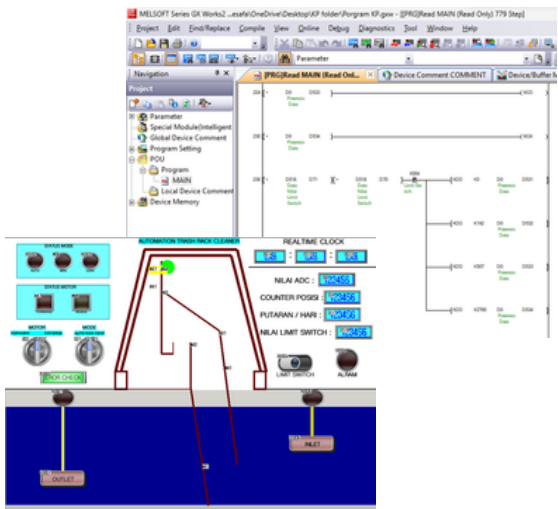
PT. Renerconsys



Desain Automatioan Trashrack Cleaner



TrashTrash
Kunci Putih Semarang

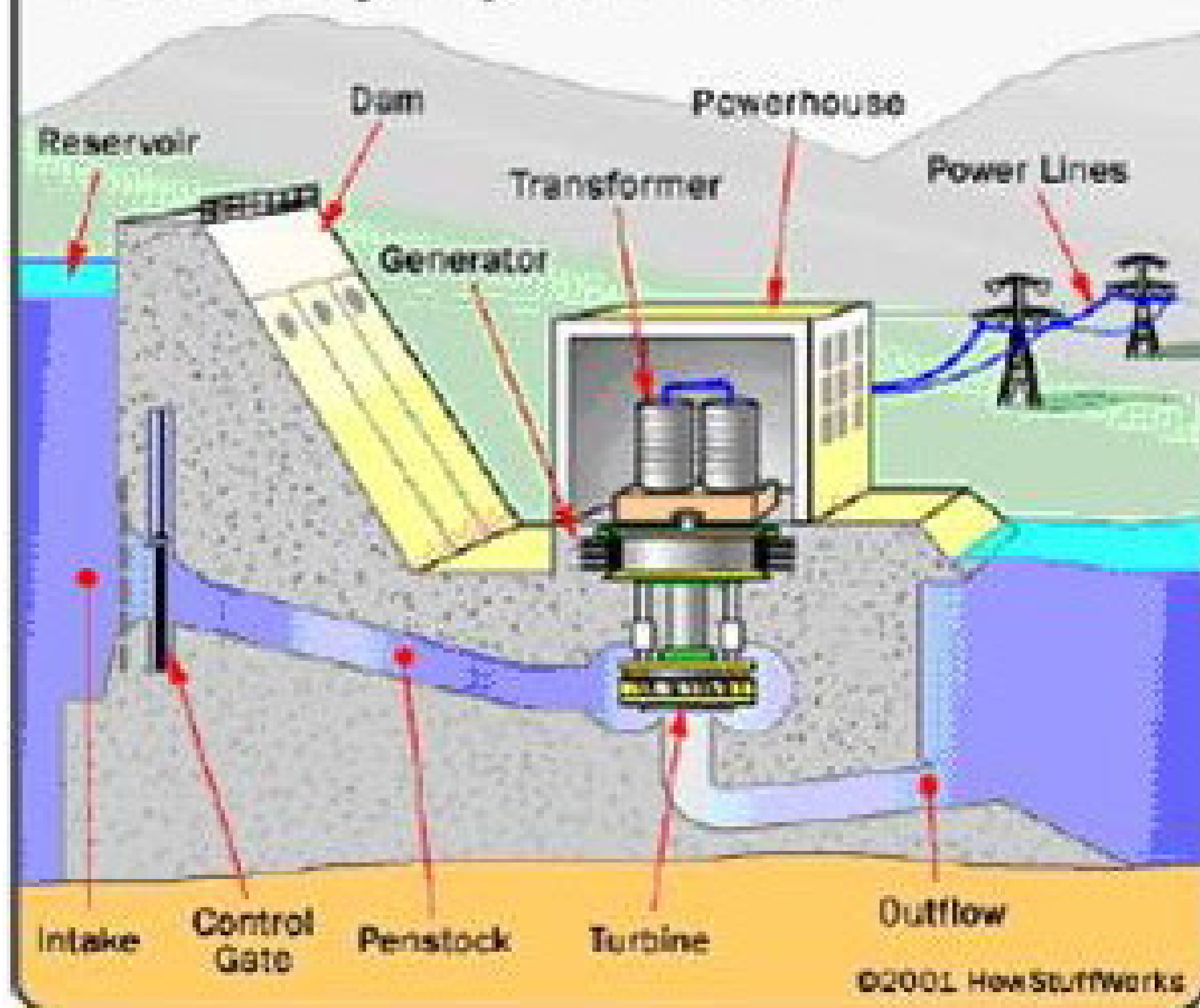


Simulasi



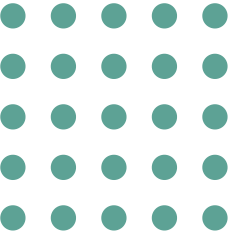
Desain Emulator PLC & HMIr

Inside a Hydropower Plant



Tujuan Kegiatan

Melakukan simulasi sistem secara full software tanpa perangkat keras, untuk mengamati respon dan perubahan arah pergerakan penggaruk



Rumusan Masalah

01. Rumusan Satu

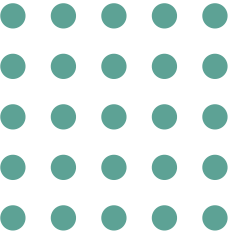
Bagaimana Pembacaan Nilai ADC potensio untuk menentukan posisi pergerakan simulasi dalam menggerakkan Penggaruk ?

02. Rumusan Dua

Bagaimana Pengguna water level dalam Mengoperasikan Automation Trashrack Cleaner ?

03. Rumusan Tiga

Apa saja respon sistem terhadap perubahan kondisi saat disimulasikan secara virtual?



Batasan Masalah

01. Batasan Satu

Sistem operasi perangkat lunak untuk pemograman PLC menggunakan GX works 2

02. Batasan Dua

Perancangan Visualisasi HMI menggunakan Perangkat GT Designer 3

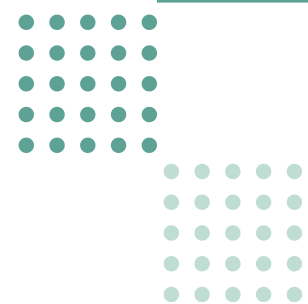
03. Batasan Tiga

Simulasi hanya dilakukan tanpa melibatkan perangkat keras fisik seperti PLC, sensor, dan motor.

04. Batasan Empat

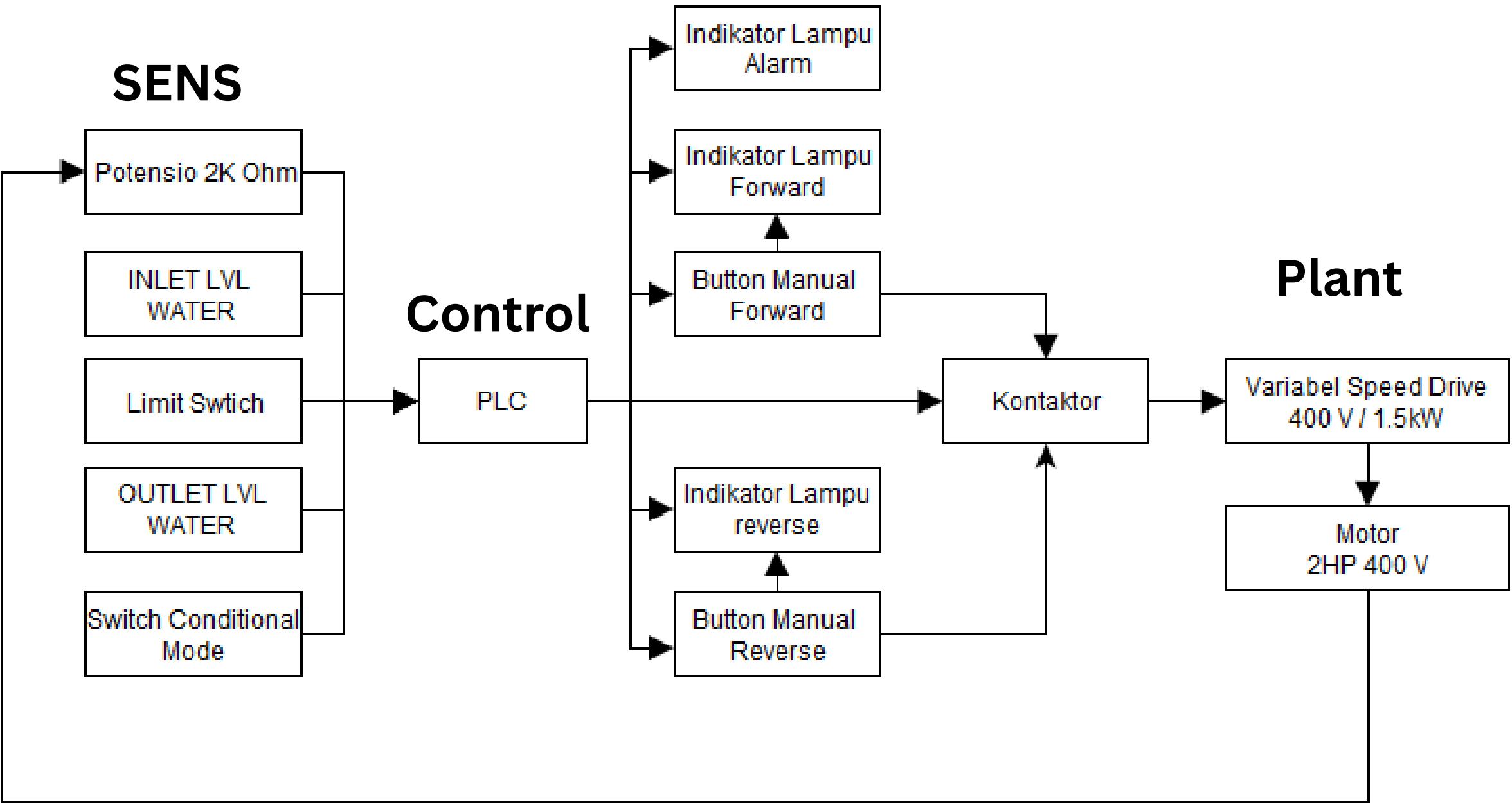
Sistem yang disimulasikan adalah pergerakan penggaruk pada trashrack

METODOLOGI KERJA PRAKTEK

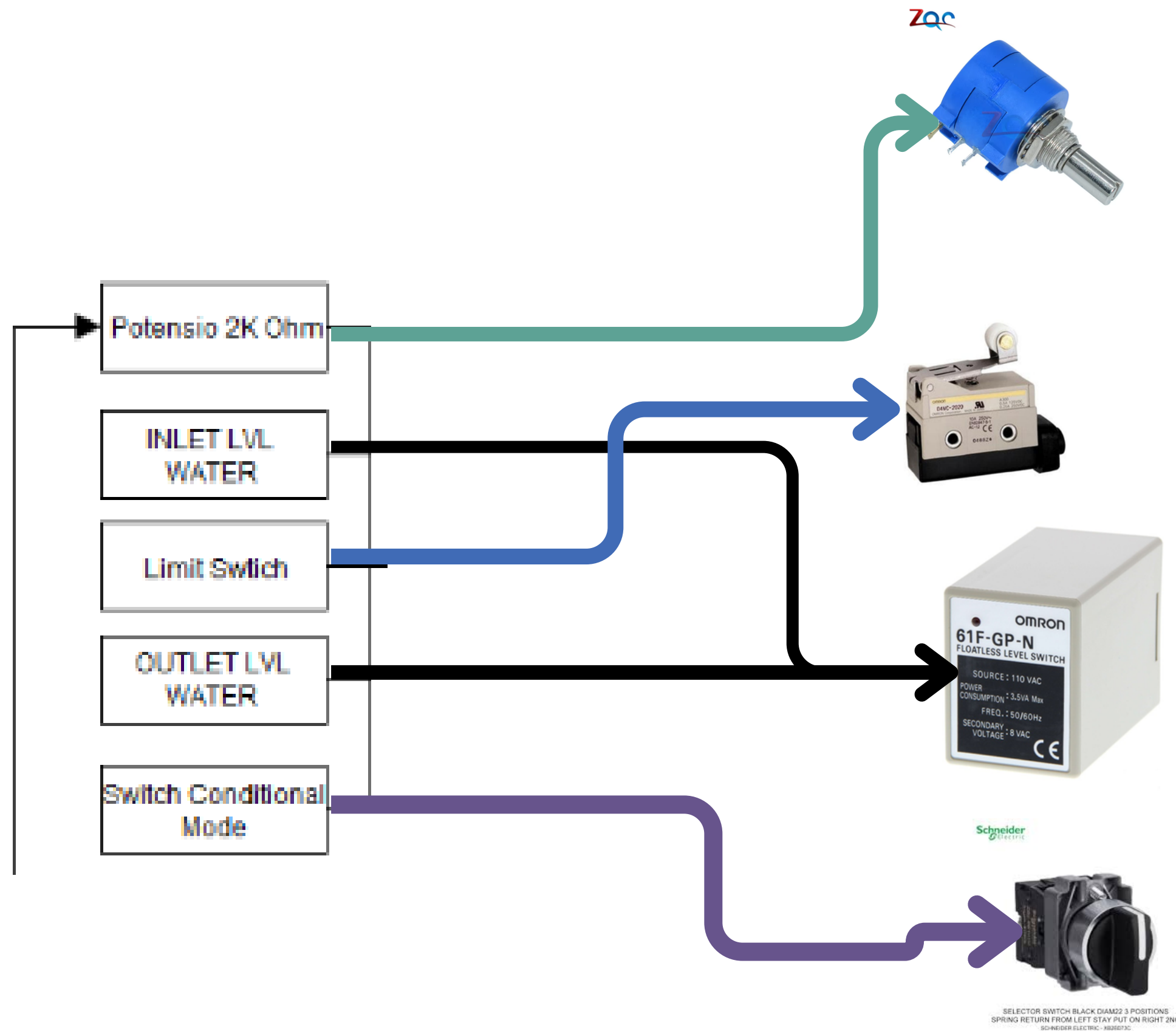


Deskripsi Automation Trash Rack Cleaner

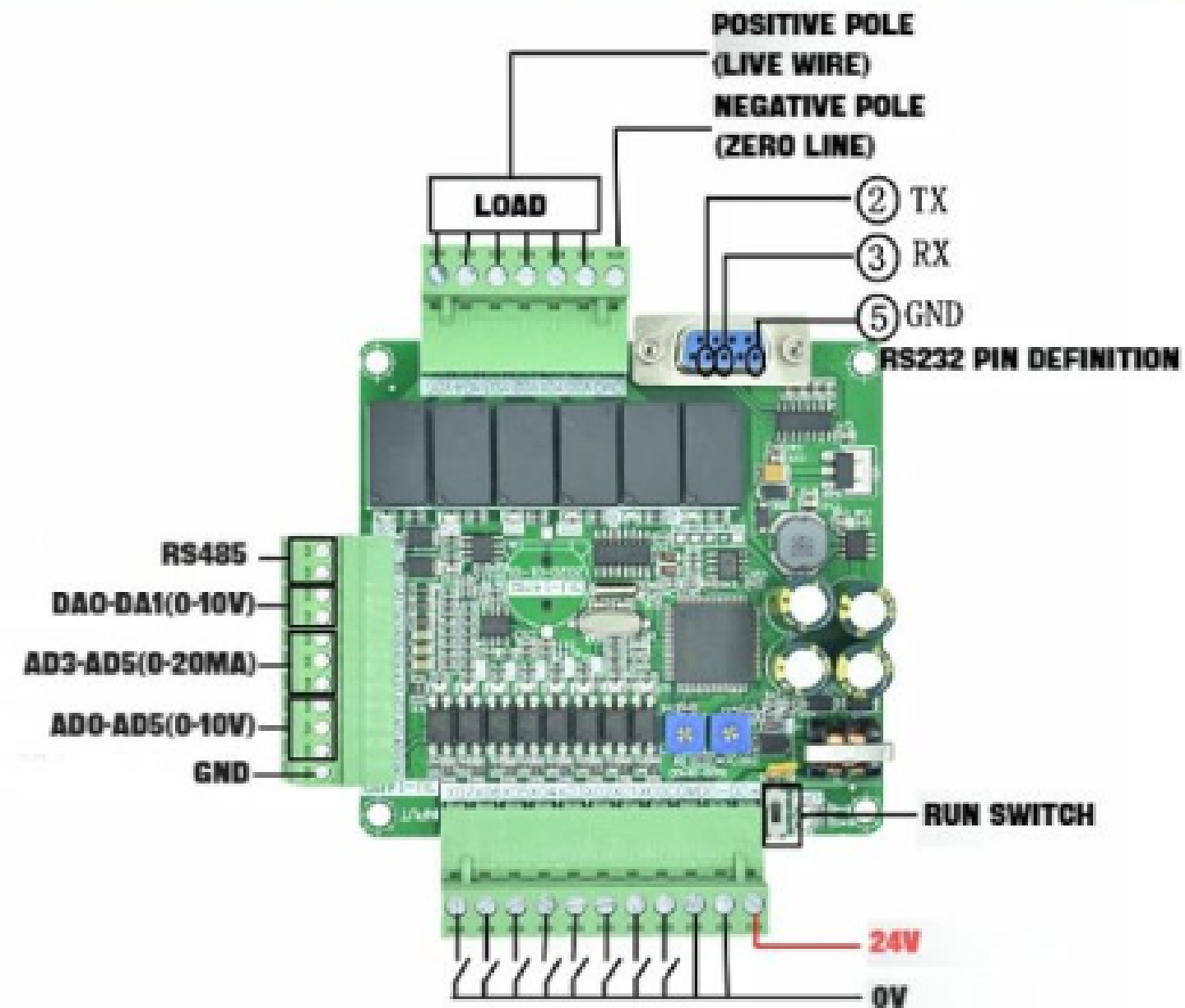
Diagram Block Circuit



Sens



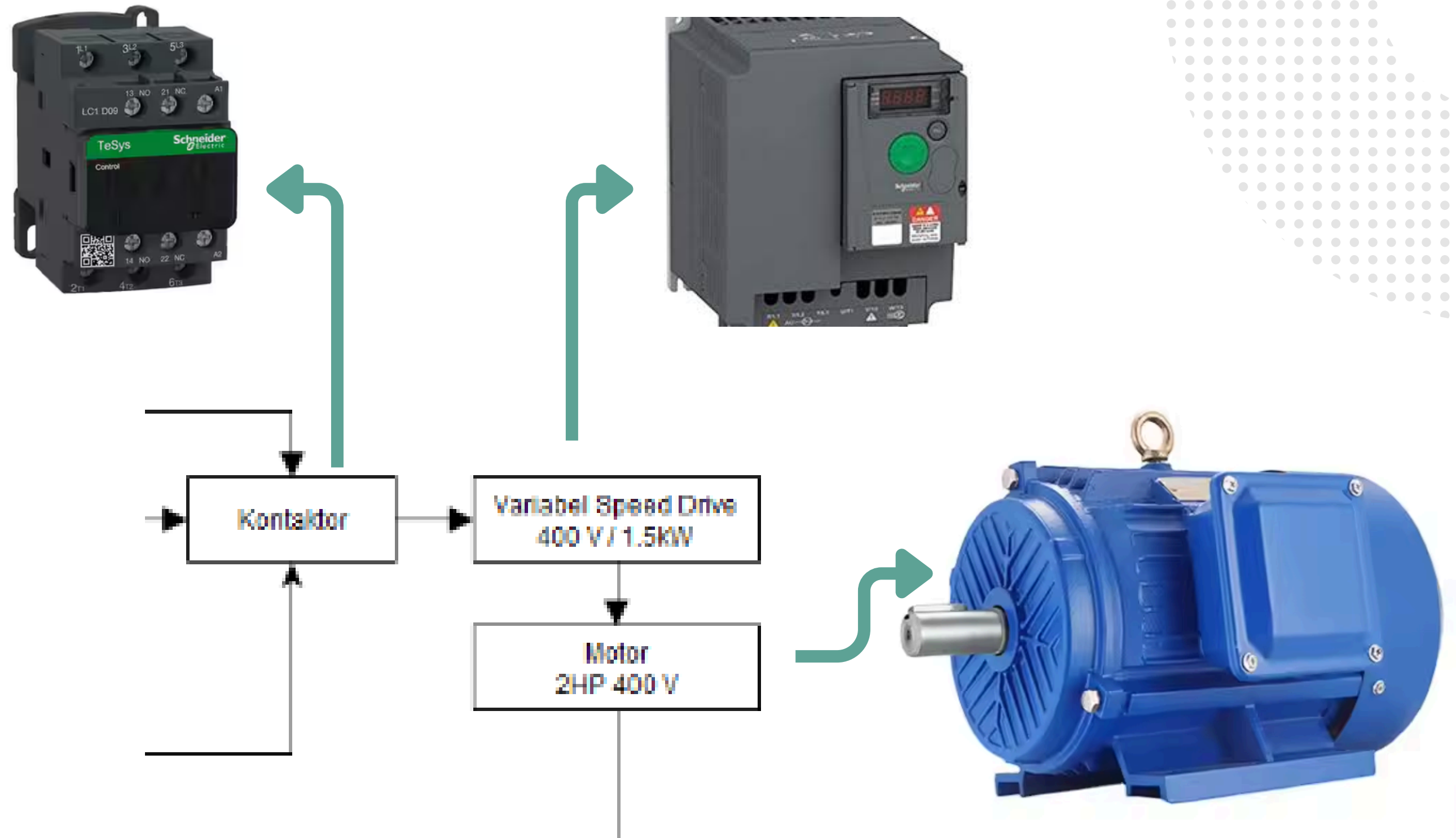
Controller



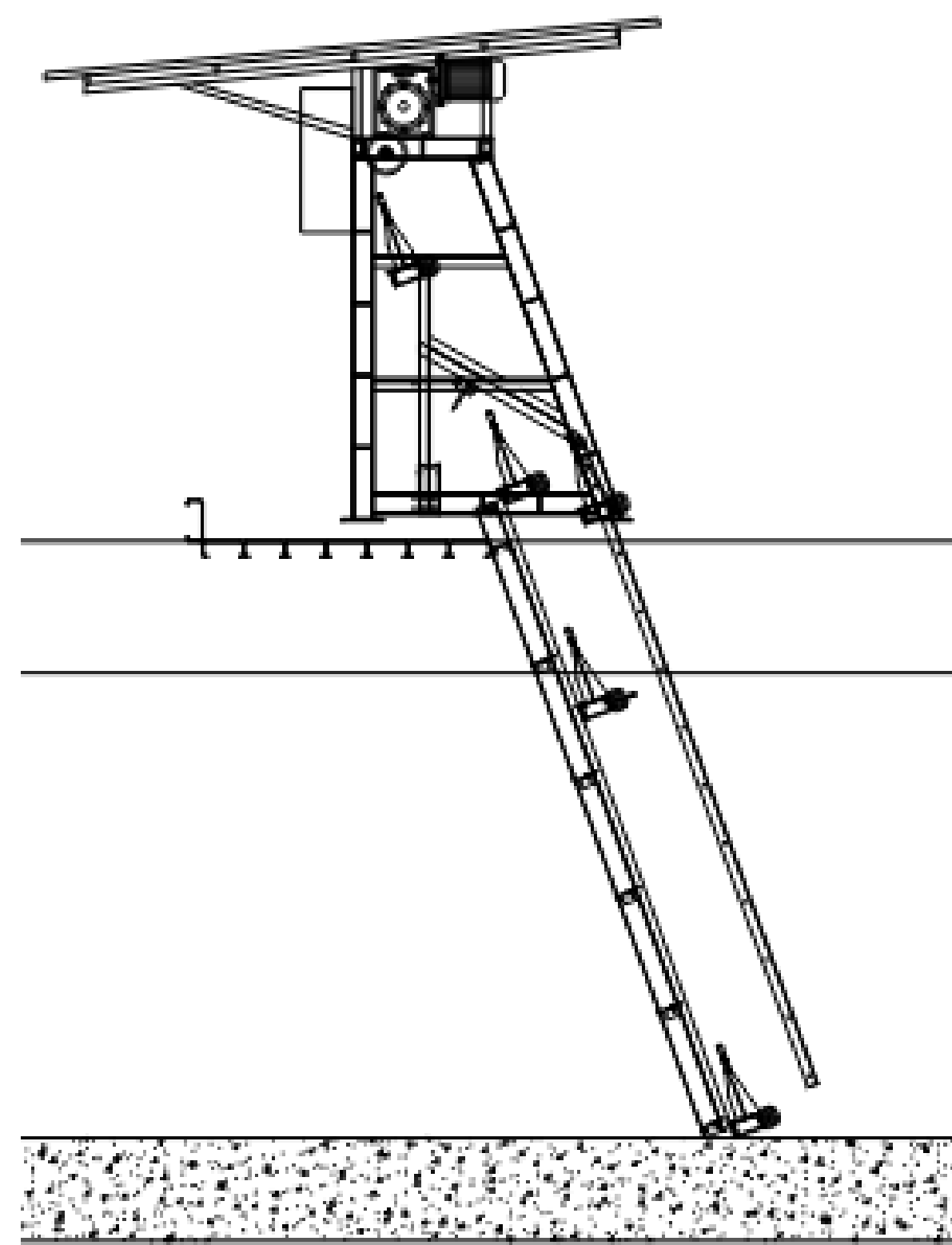
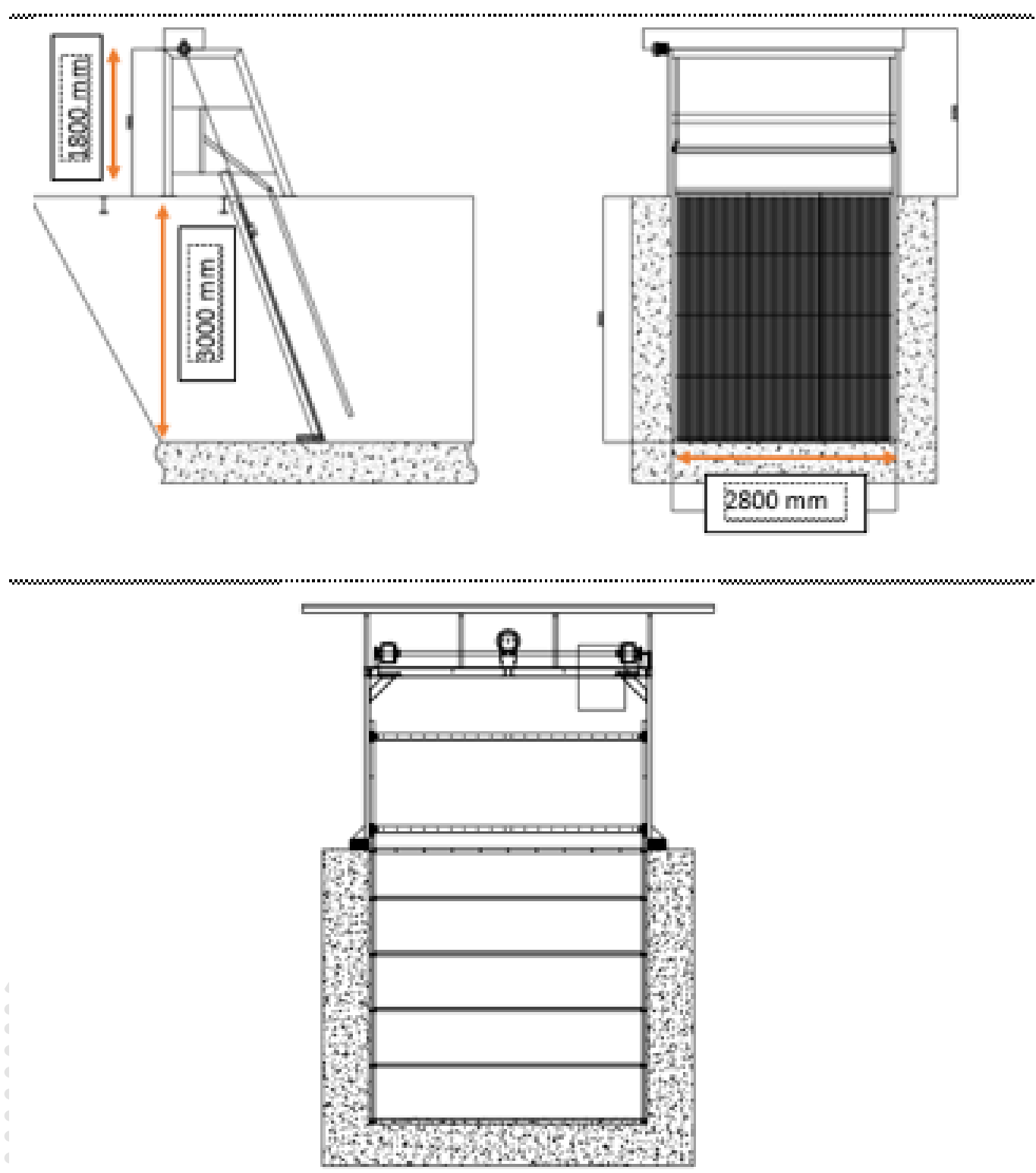
Mitsubishi FX3U-14MR

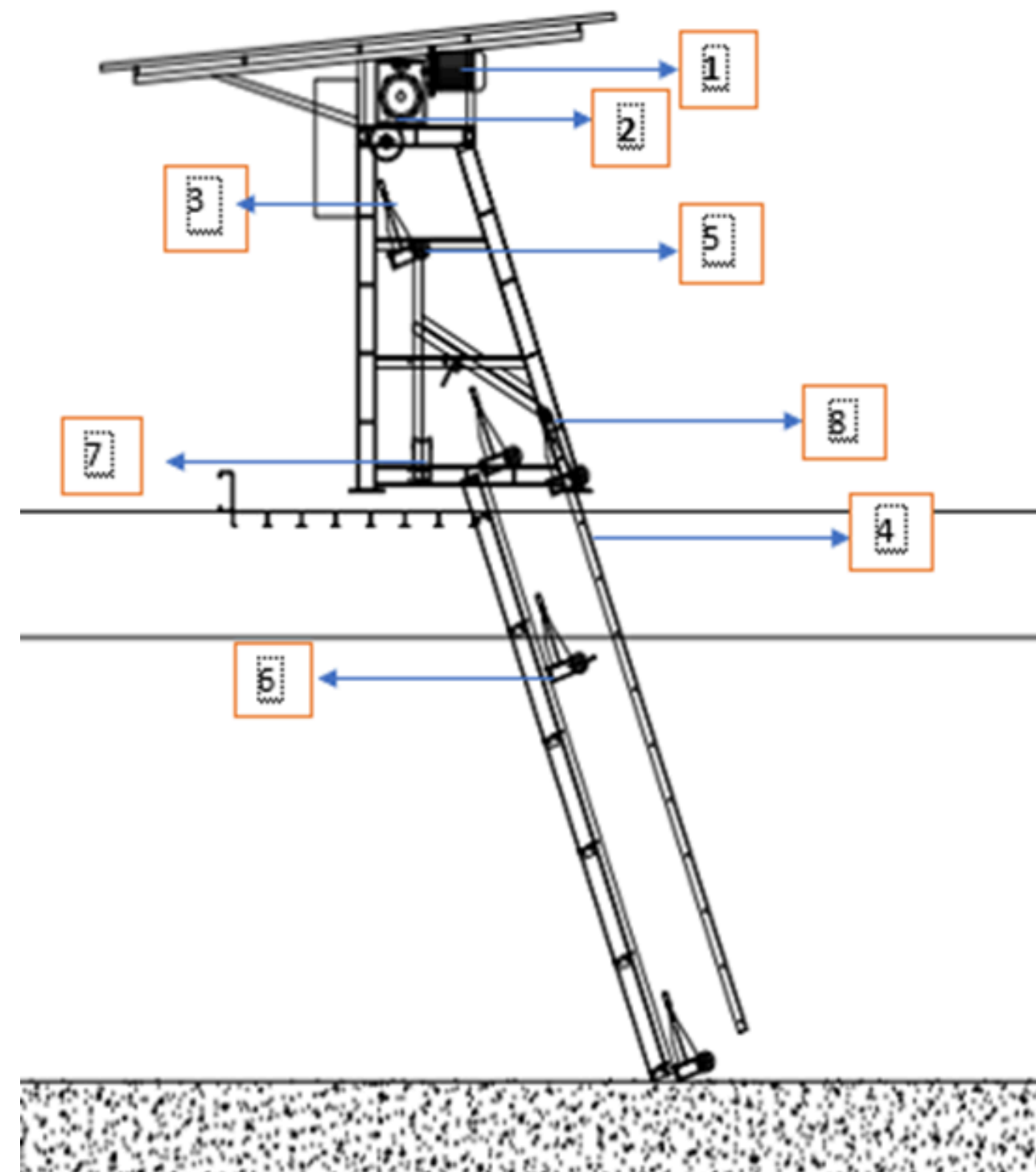
- Power Supply 24 VDC
- Brand: Lollette
- Model: LE3U 14MR6AD2DA
- Supply voltage: 24 VDC
- Digital inputs: 8
- Digital outputs: 6
- Type of outputs: relay
- Output capacity: 5A @ 250 VAC
- Analog ports: 6
- Communication ports: RS232, Communication ports: RS485
- Supports HMI: Yes
- HMI compatible model: OP320-A
- Supports Encoder: Yes
- MCU: 32-bit
- Speed: 3 kHz
- Program capacity: 8,000 lines of LADDER
- DIN rail mounting: Yes

Plant



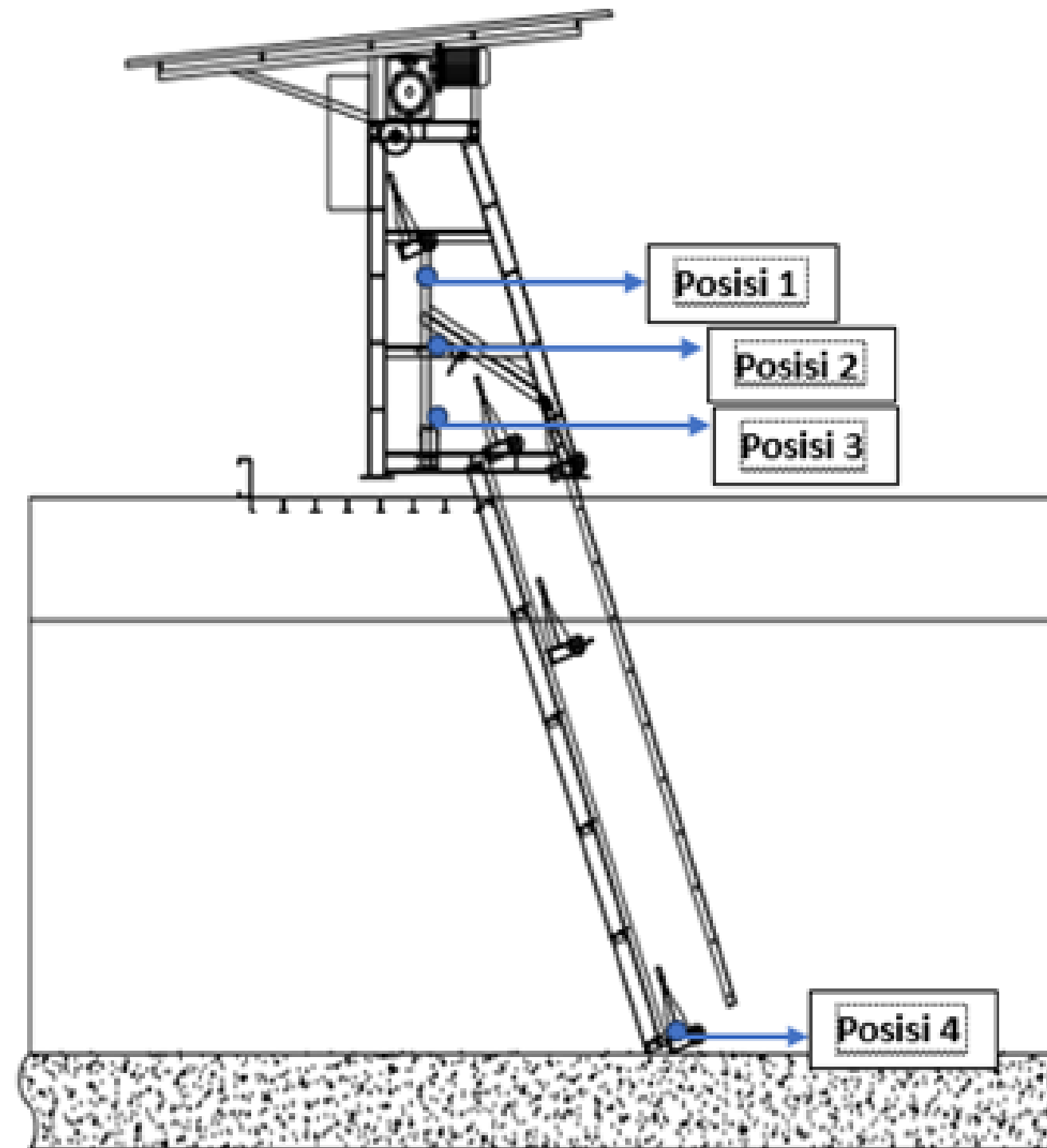
Desain Fisik dan Dimensi Sistem





1. Motor Penggerak
2. Drum Tali Kawat
3. Tali Penggerak
4. Rel Pengarah
5. Roda Penggaruk
6. Penggaruk Sampah
7. Tempat Pembuangan Sampah
8. Engsel Pengembalian

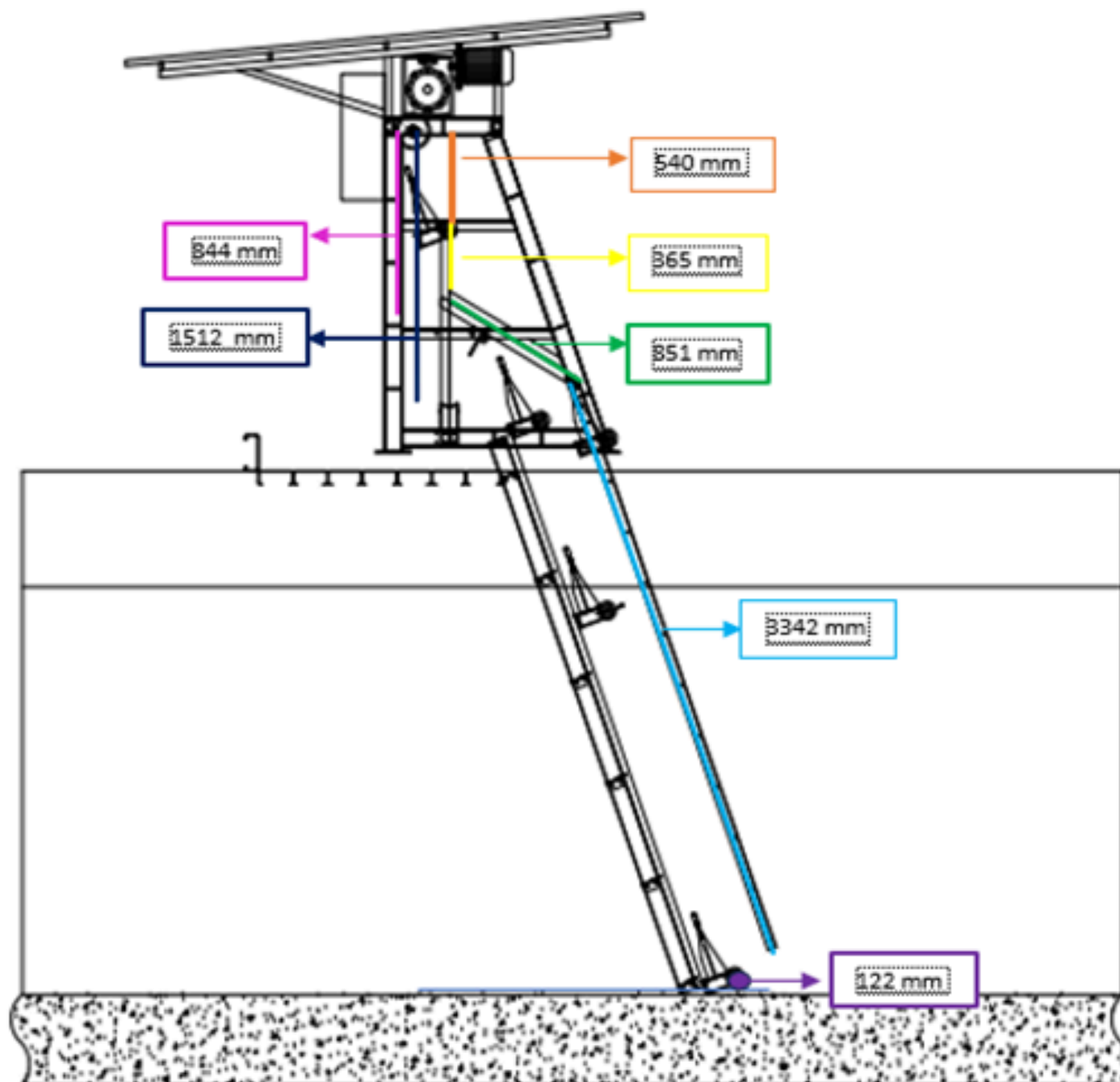
Alur Pegerakan Penggaruk



Parameter Pada Simulasi Automation Trash Rack Cleaner

1. Pembacaan Senesor Nilai Tegangan Pada PLC : 10V ADC 12 bit input alamat D0
2. Status Sensor Inlet dan outlet : input Inlet alamat X006 , input Outlet alamat X007 aktive High
3. Input selector switch : Swtich arah ke kiri input alamat X002 (mode auto), Swtich arah ke kanan input alamat X003 (continue)

4. Insialisasi referensi posisi

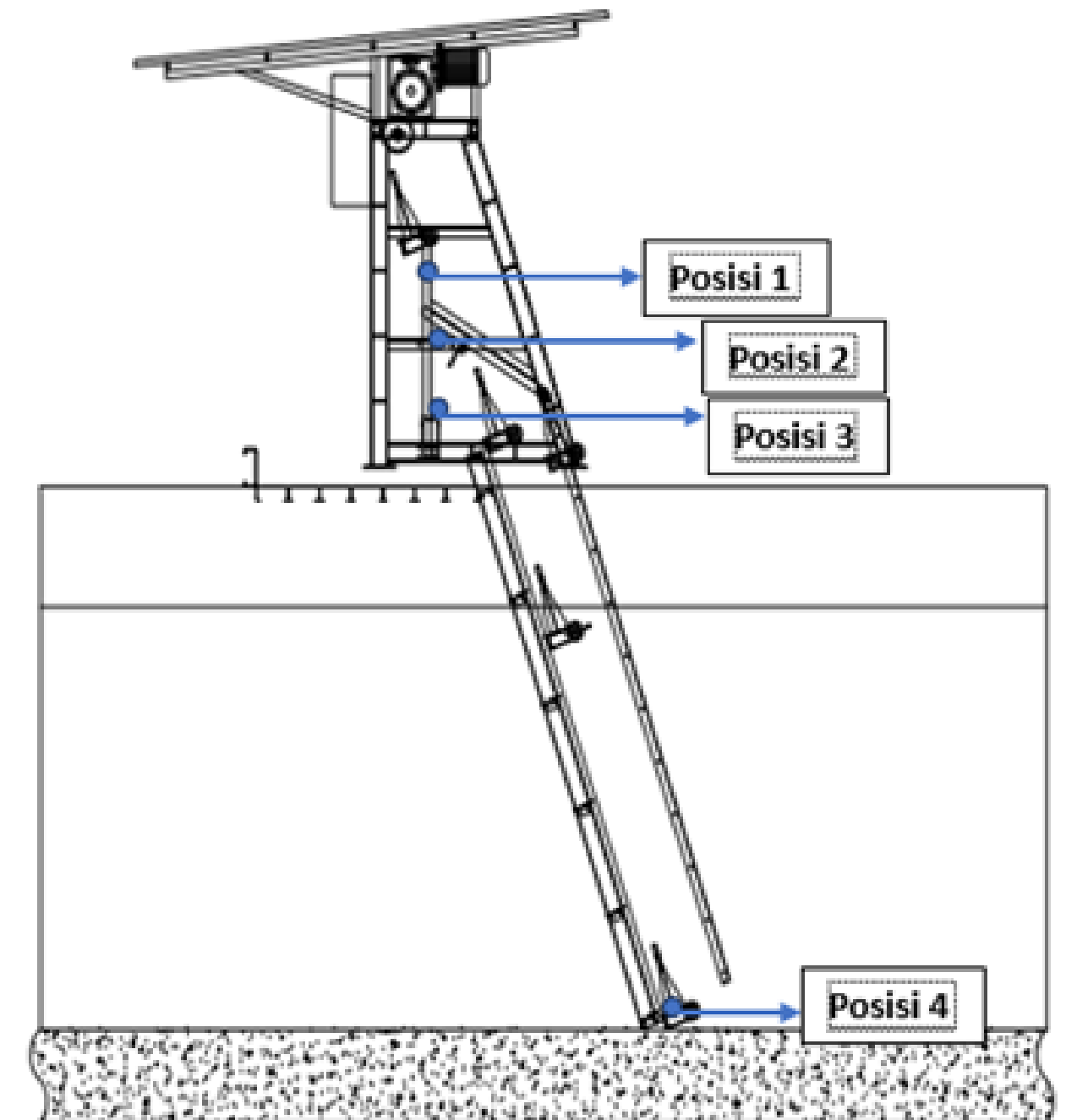


- a) Warna oranye: Jarak dari drum ke limit switch, yaitu 540 mm.
- b) Warna kuning: Panjang rangka bagian 1, yaitu 365 mm.
- c) Warna hijau muda: Panjang rangka bagian 2, yaitu 851 mm.
- d) Warna biru muda: Panjang rangka bagian 3, yaitu 3342 mm.
- e) Warna ungu: Diameter roda pada bagian penggaruk, yaitu 144 mm.
- f) Warna biru tua: Jarak dari drum ke bagian pembuangan sampah (posisi 3), yaitu 1512 mm.
- g) Warna merah muda (pink): Jarak posisi 2 dari drum, yaitu 955 mm.

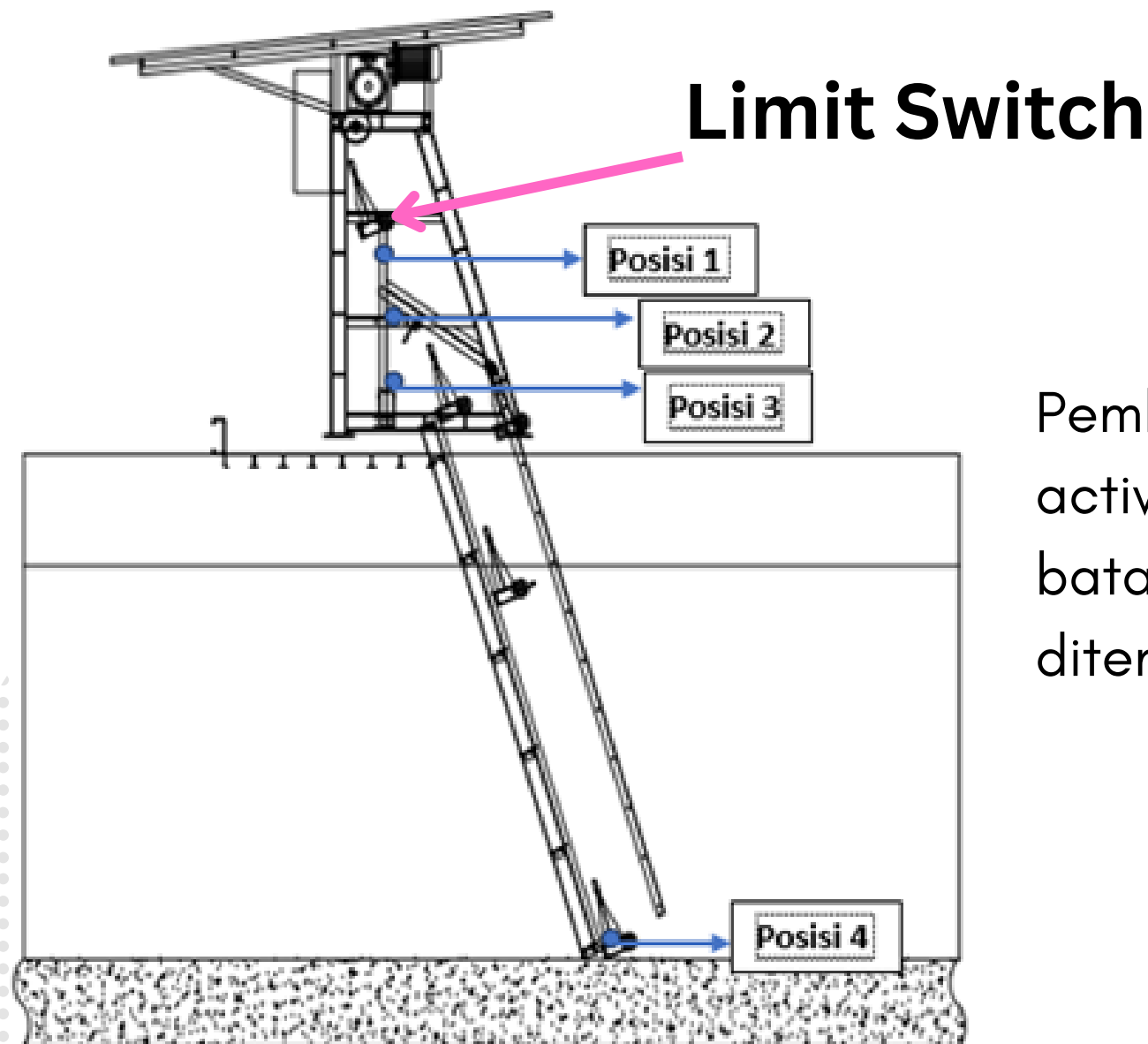
Parameter yang akan digunakan pada Pada Penggerak Penggaruk Automatin Trashrack Cleaner

masing jarak pada setiap posisi yang akan digunakan sebagai parameter:

- a) Posisi Pemberhentian Penggaruk Posisi 1 Jarak: 674 mm, Tegangan: 1 V, Nilai ADC: 410
 - b) Posisi Pemberhentian Penggaruk Posisi 2 Jarak: 955 mm, Tegangan: 1,42 V, Nilai ADC: 581
 - c) Posisi Pemberhentian Penggaruk Posisi 3 Jarak: 1512 mm, Tegangan: 2,24 V, Nilai ADC: 918
 - d) Posisi Pemberhentian Penggaruk Posisi 4 Jarak: 5220 mm, Tegangan: 7,74 V, Nilai ADC: 3172
 - e) Limit switch Jarak: 540 mm, Tegangan: 0,8 V, Nilai ADC: 328
- Batas bawah Jarak: 5393 mm, Tegangan: 8 V, Nilai ADC: 3273

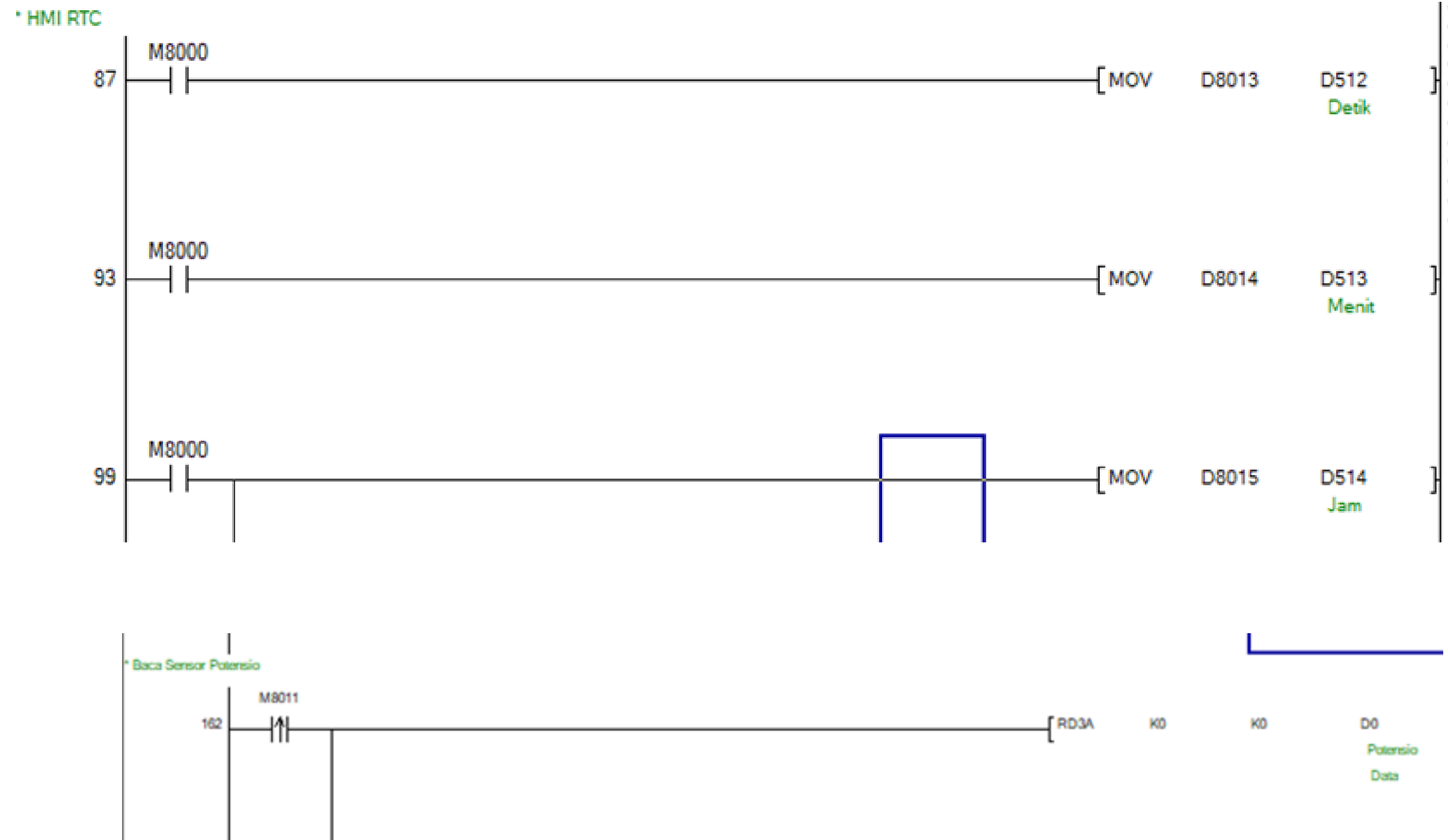
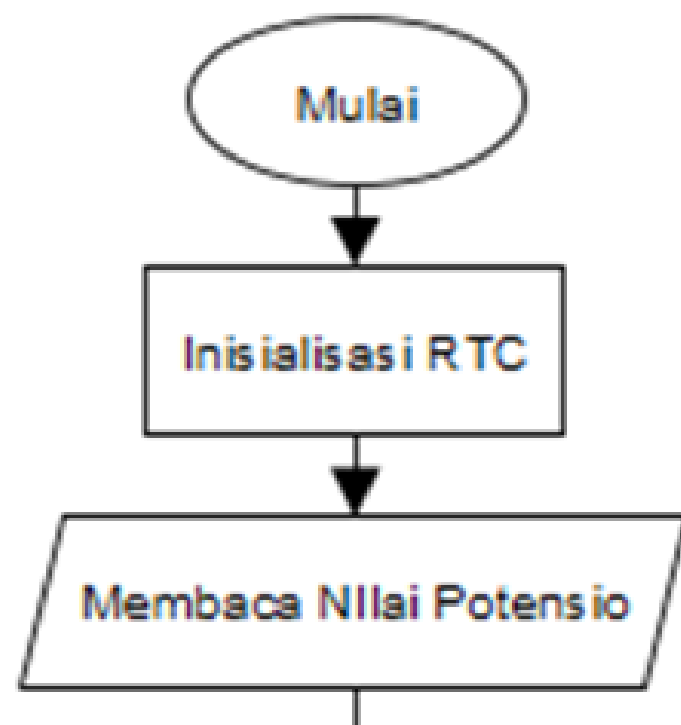


5. Limit Switch

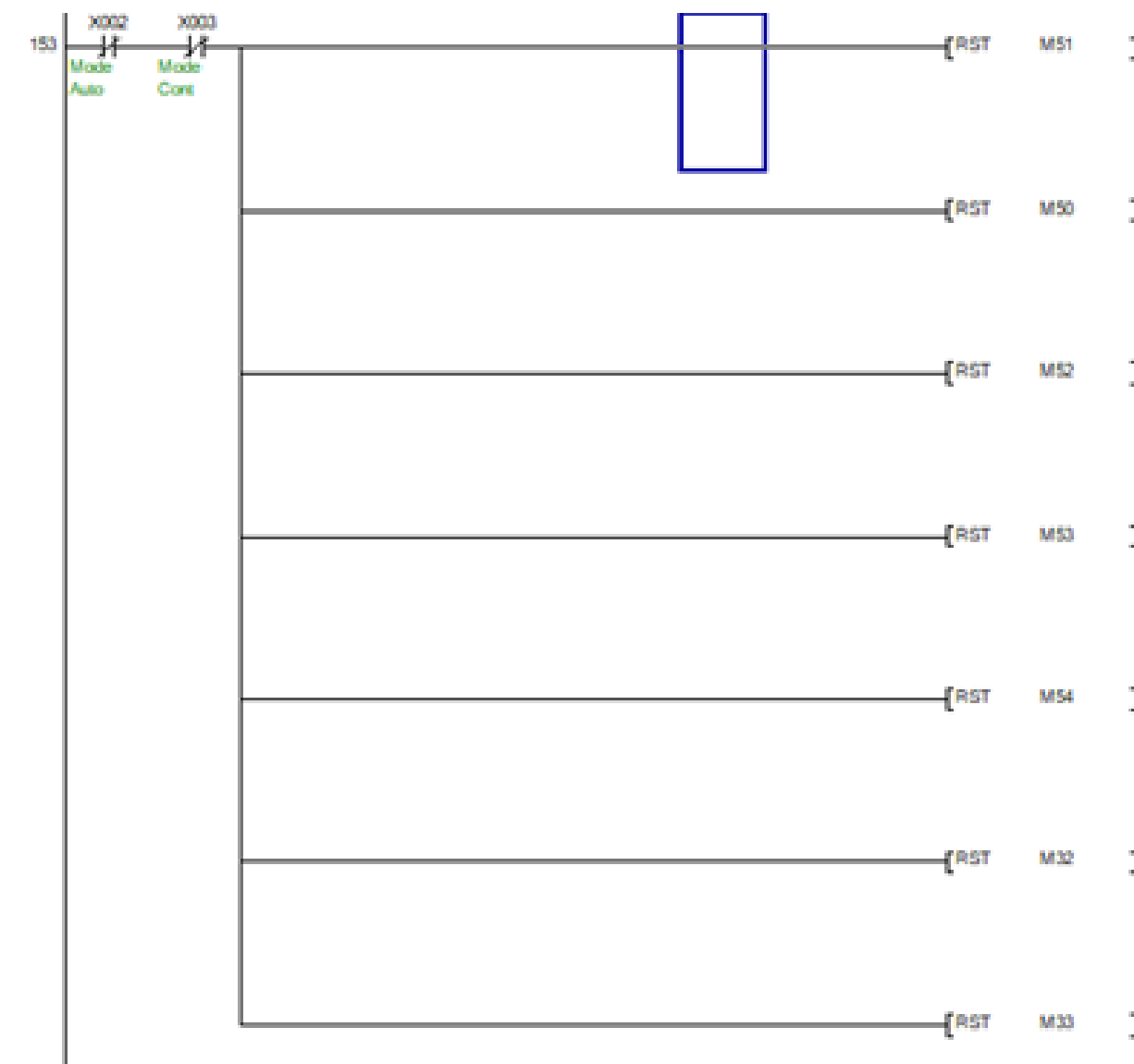
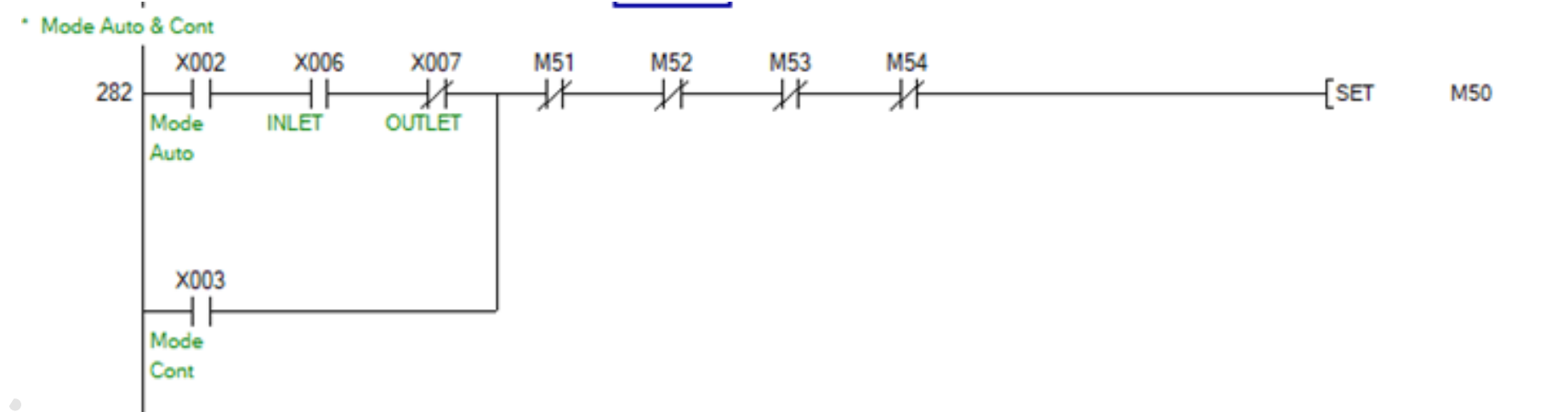
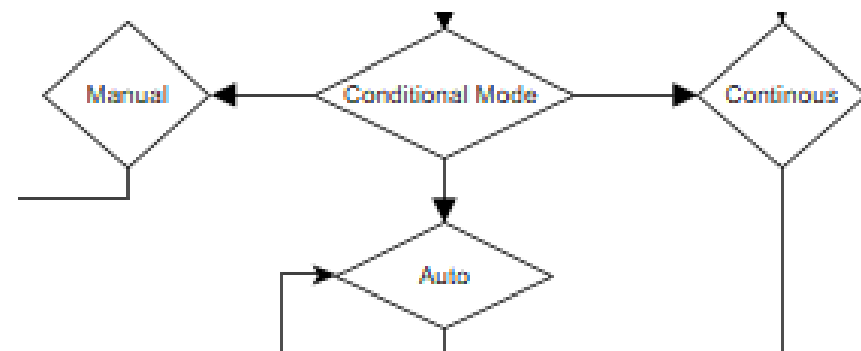


Pembacaan Status Input limit switch : Pembacaan melalui input X004 dengan active High Limit switch digunakan untuk memastikan nilai posisi tidak melewati batas yang telah ditentukan. Jika limit switch melebihi batas yang telah ditentukan, maka sistem akan mendeteksi error

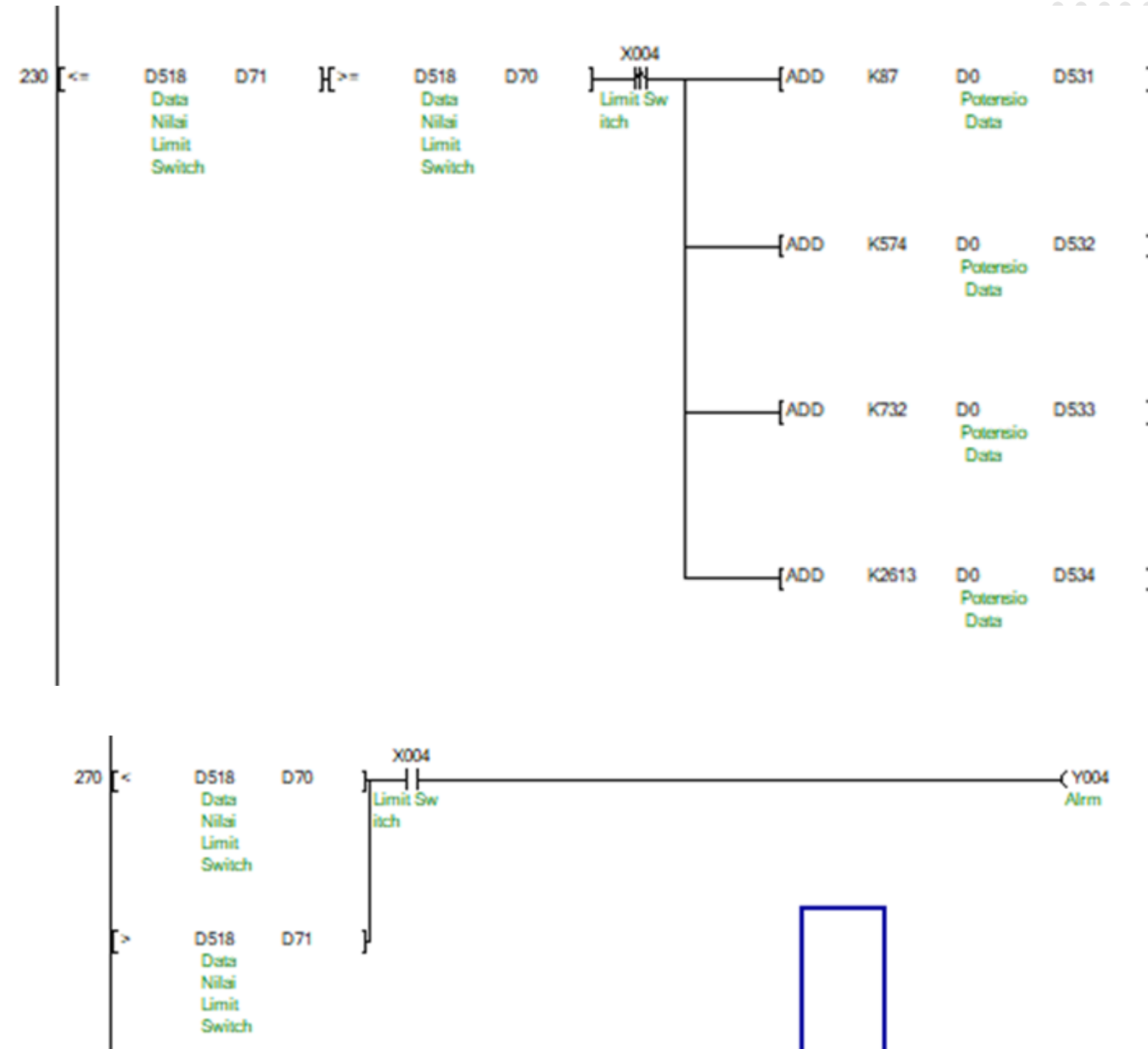
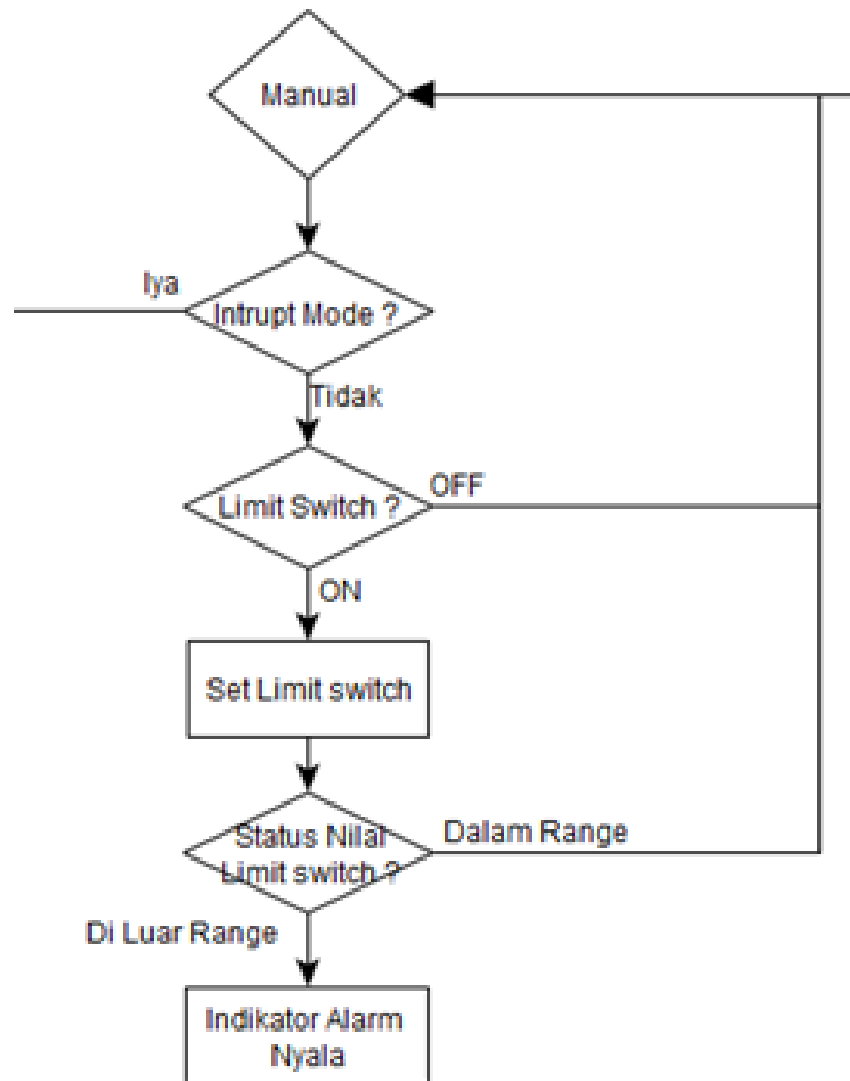
Perancangan Ladder Diagram Inisialisasi RTC dan Nilai Potensio



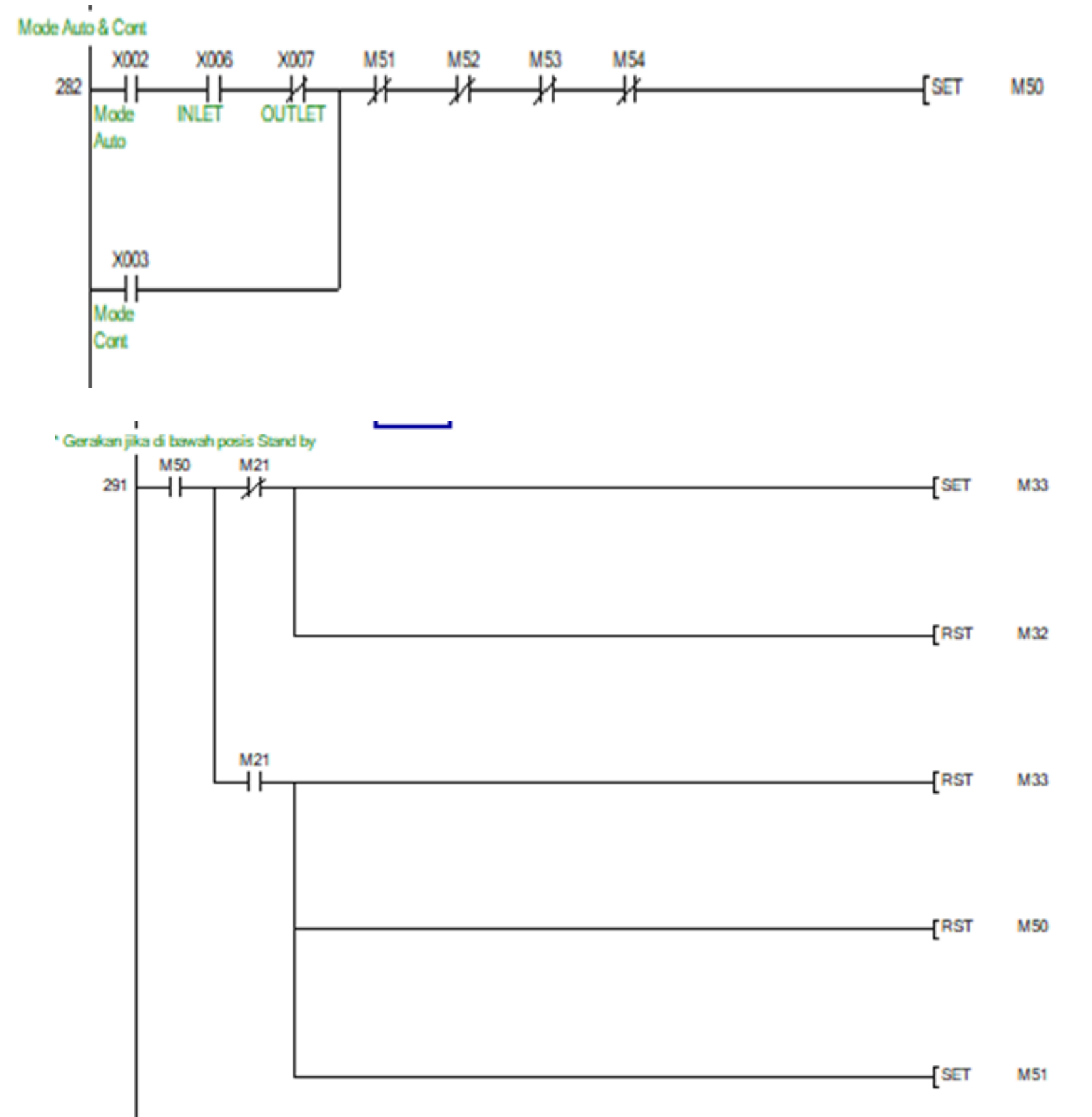
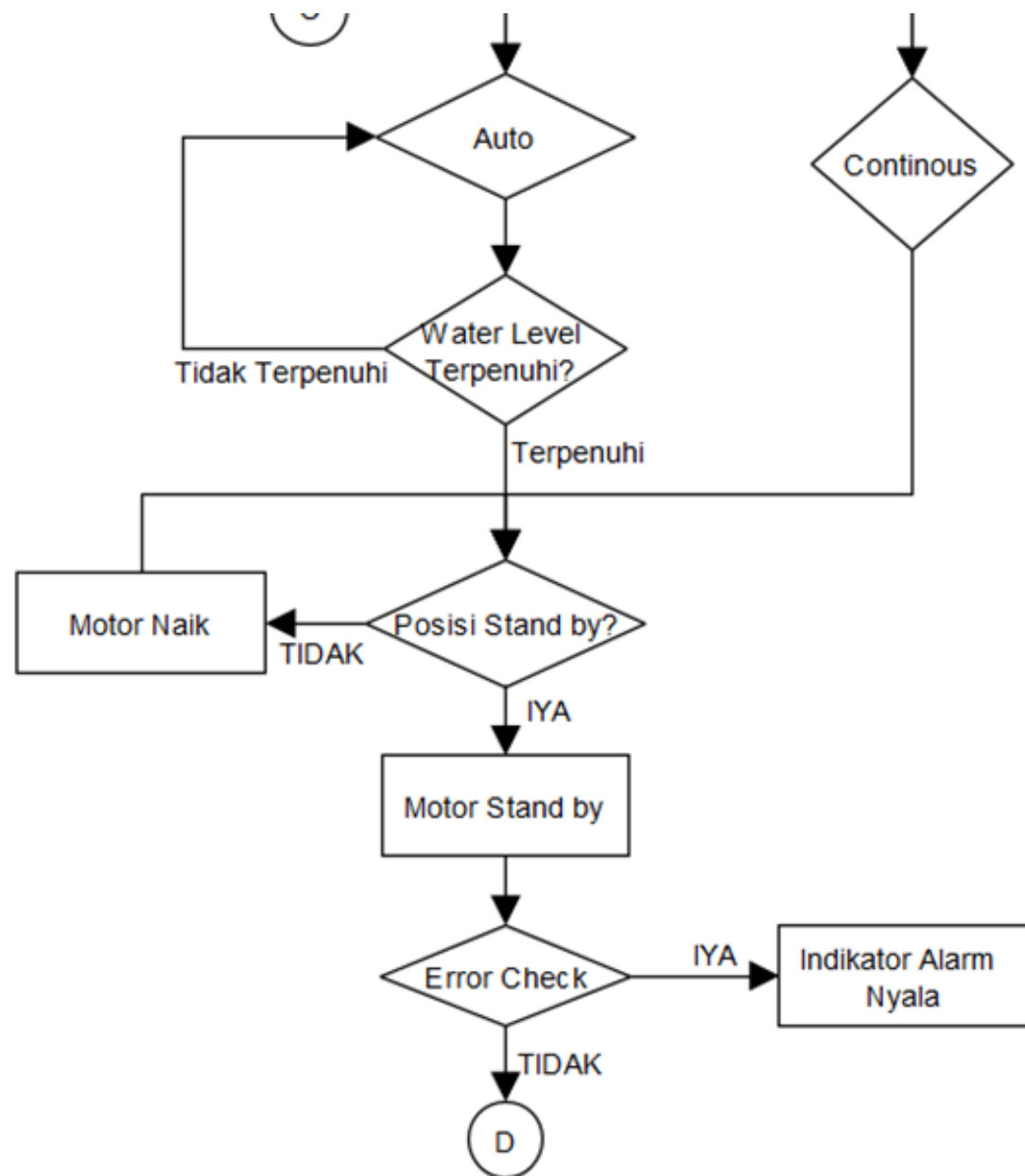
Perancangan Ladder Diagram Conditional Mode



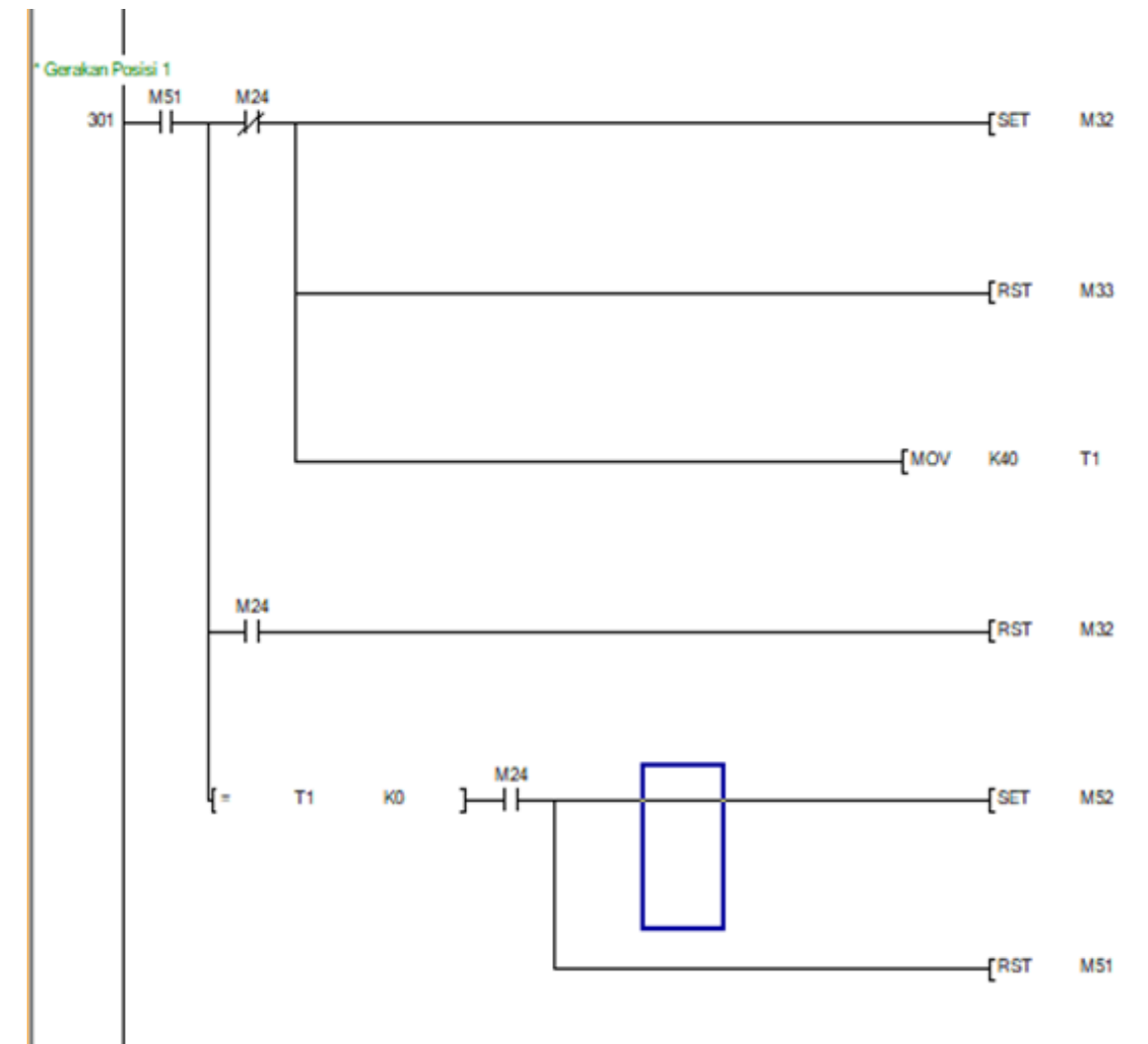
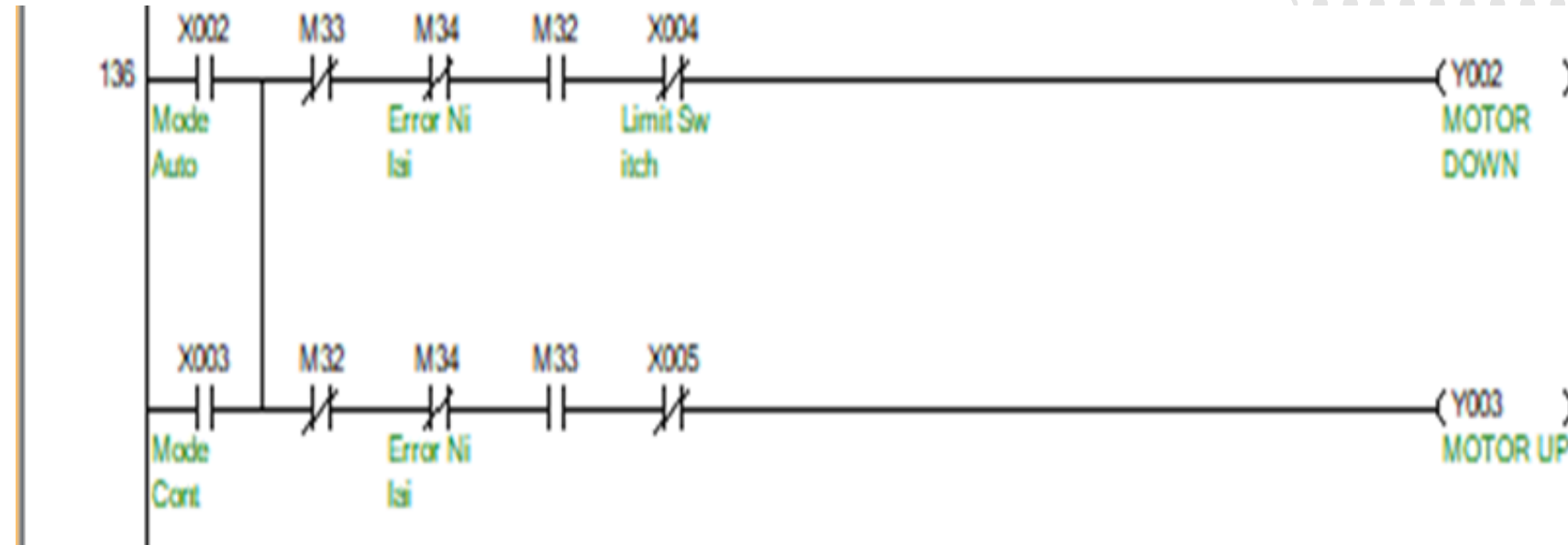
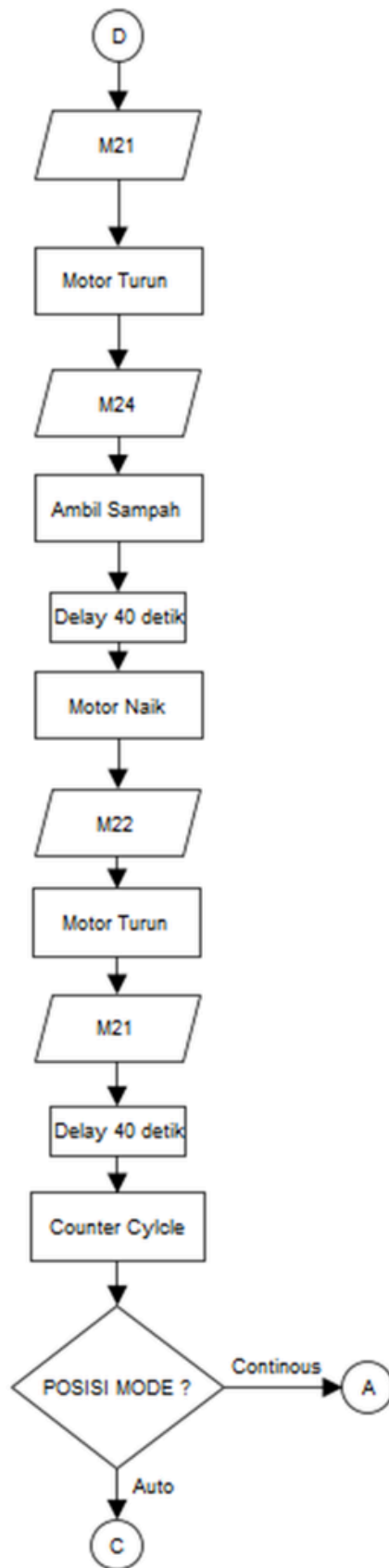
Perancangan Ladder Diagram intruksi mode manual



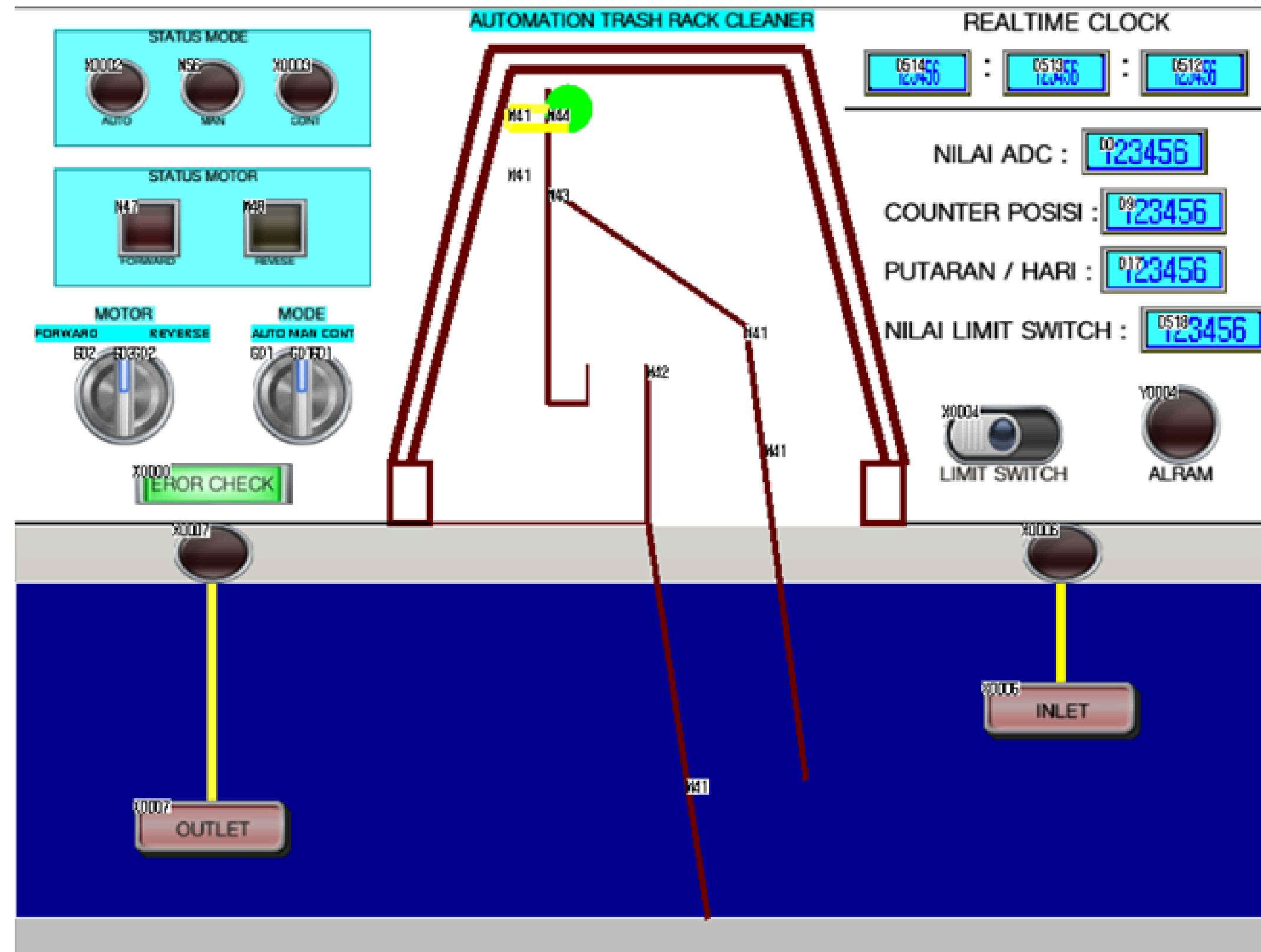
Perancangan Ladder Diagram instruksi Mode Auto



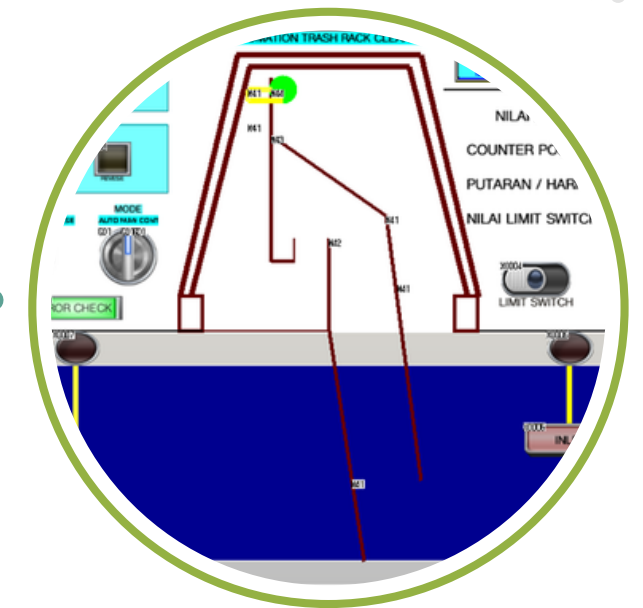
Perancangan Ladder Diagram Pergerakan Penggaruk



Perancangan HMI GT Desainer

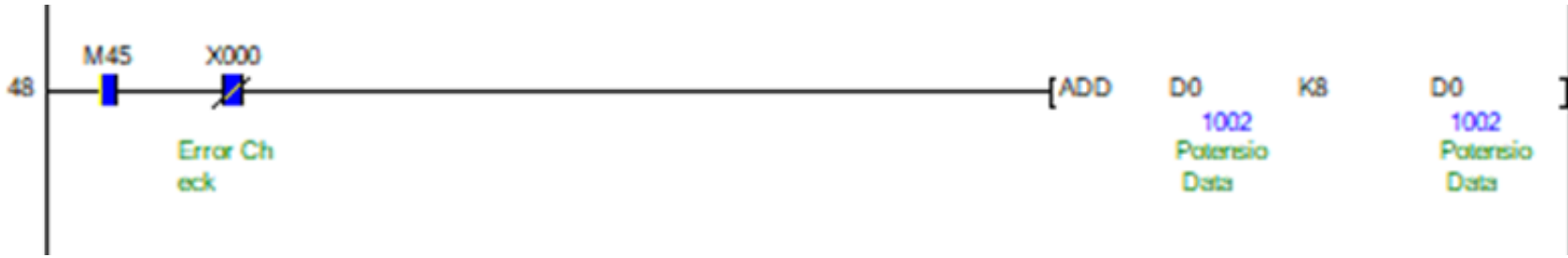
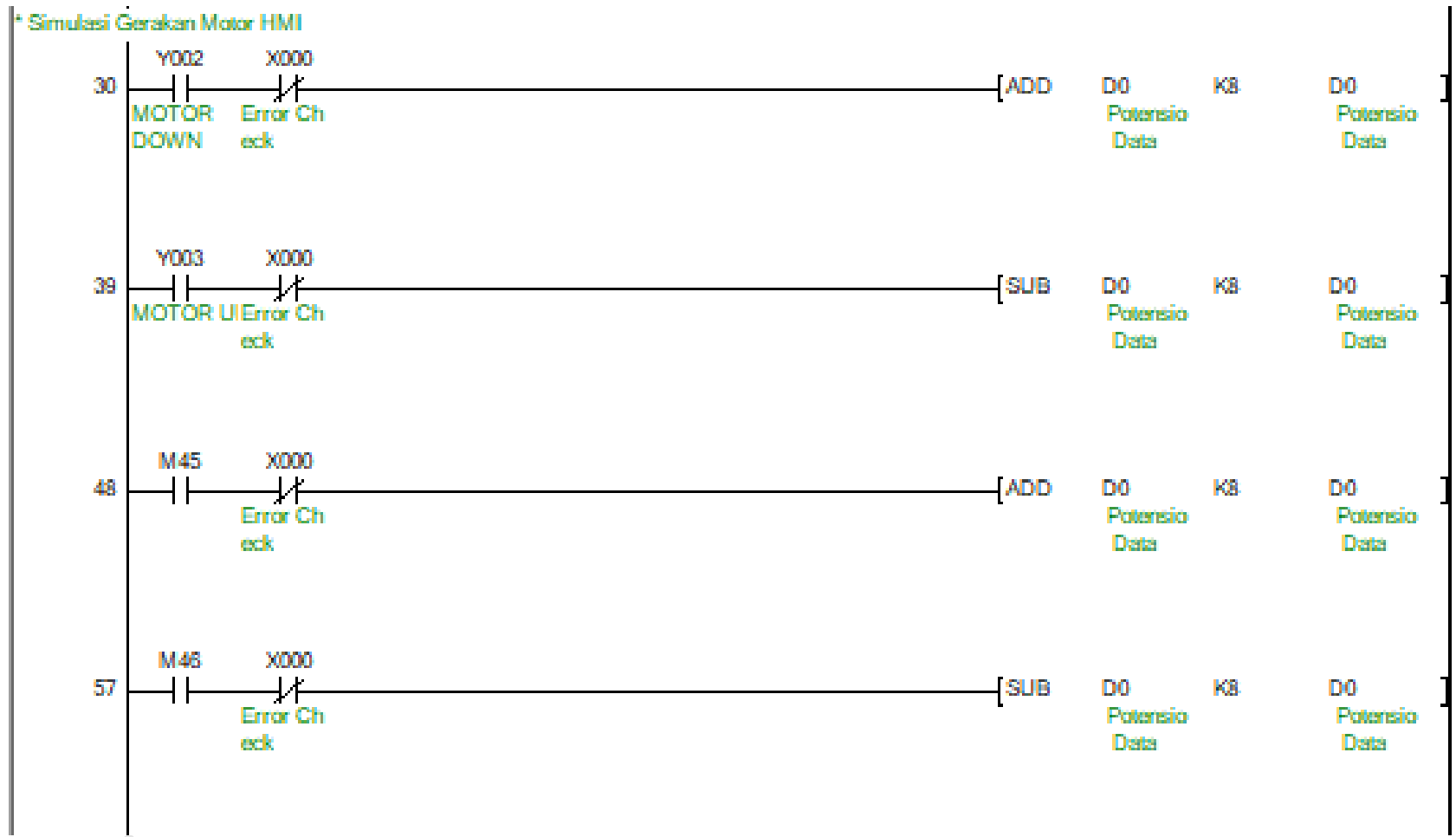


Simulasi Mode Continue

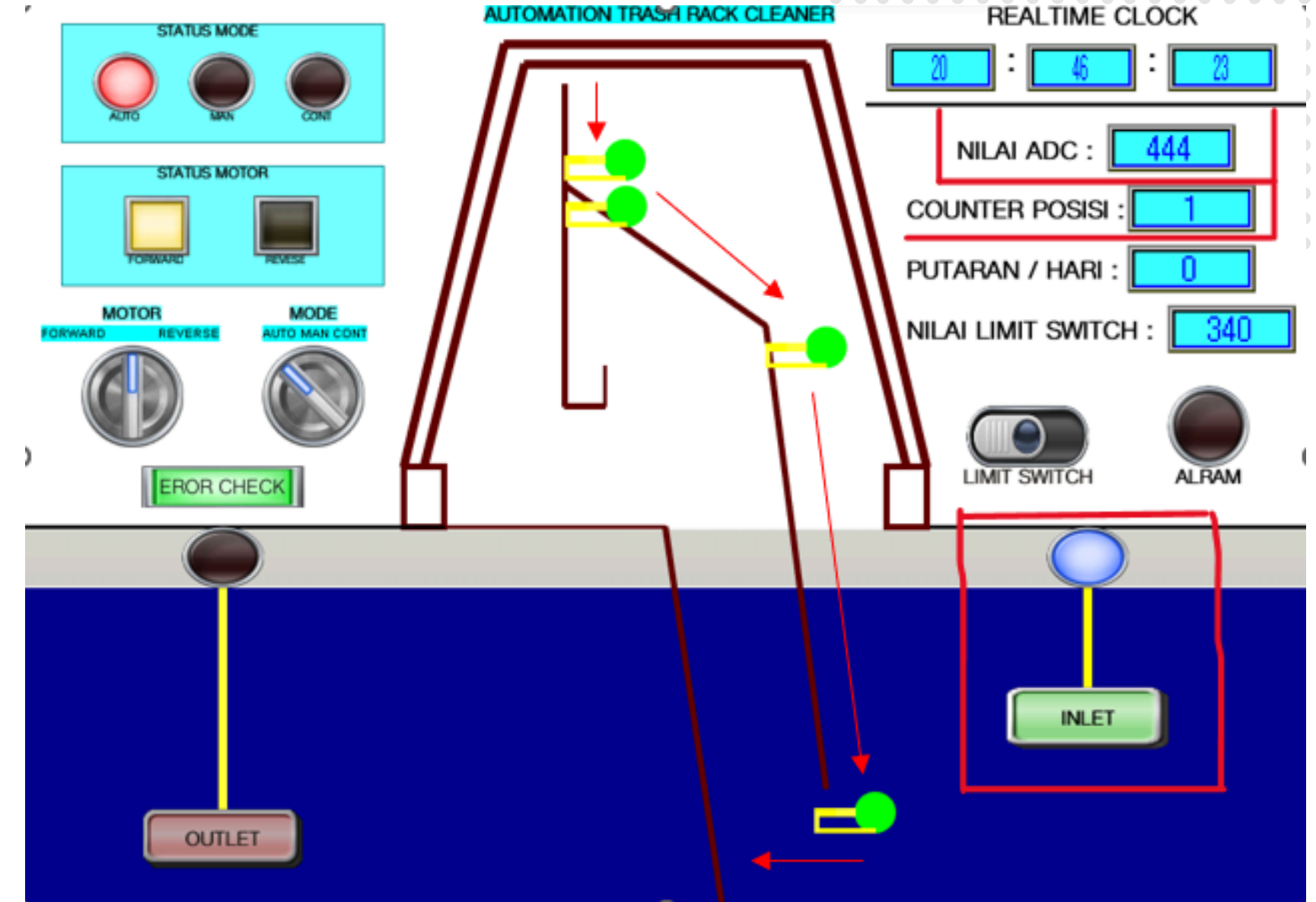
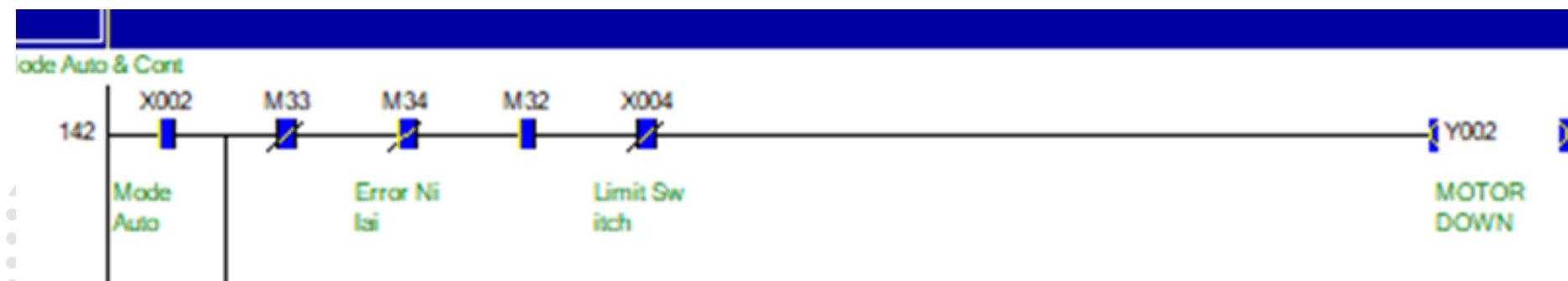
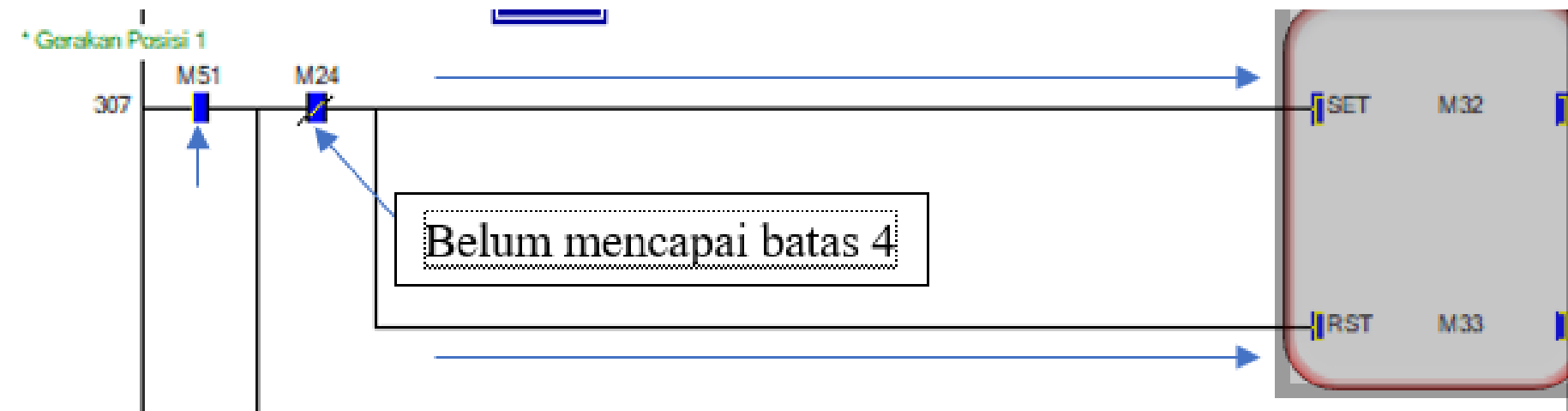
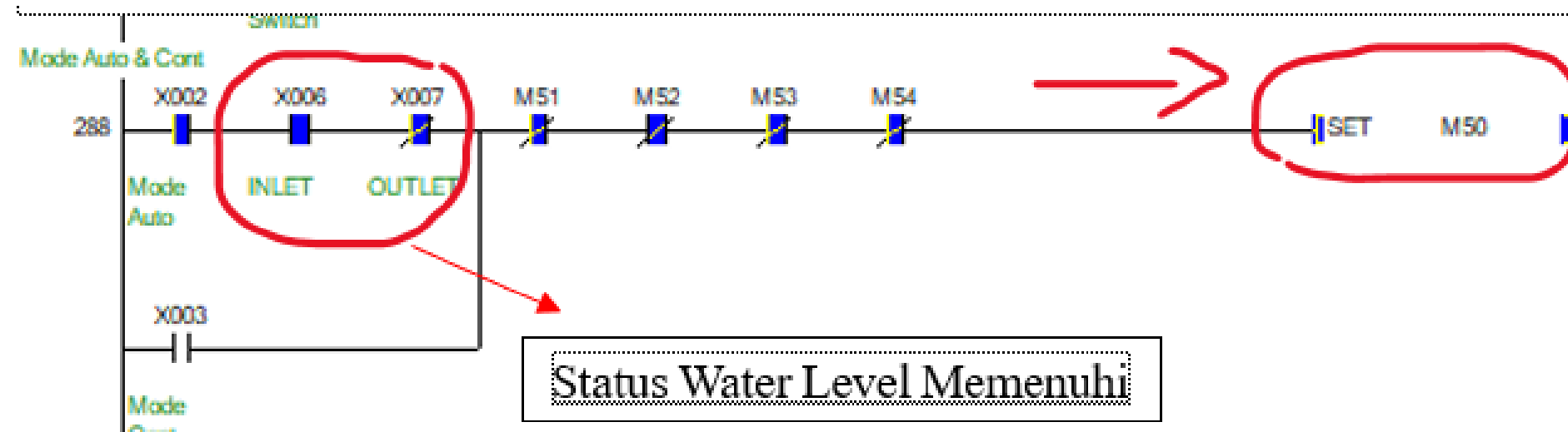


Respon HMI

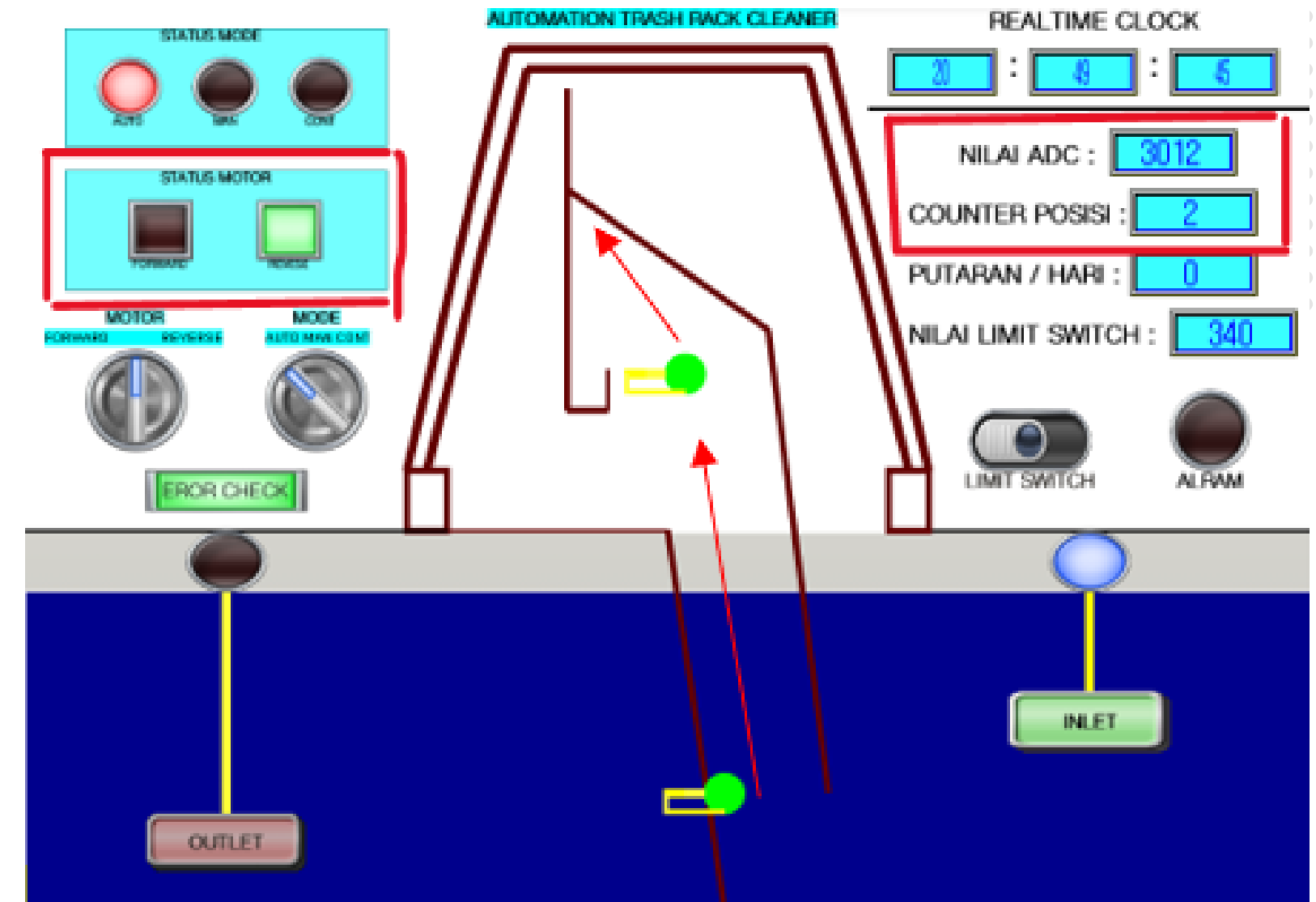
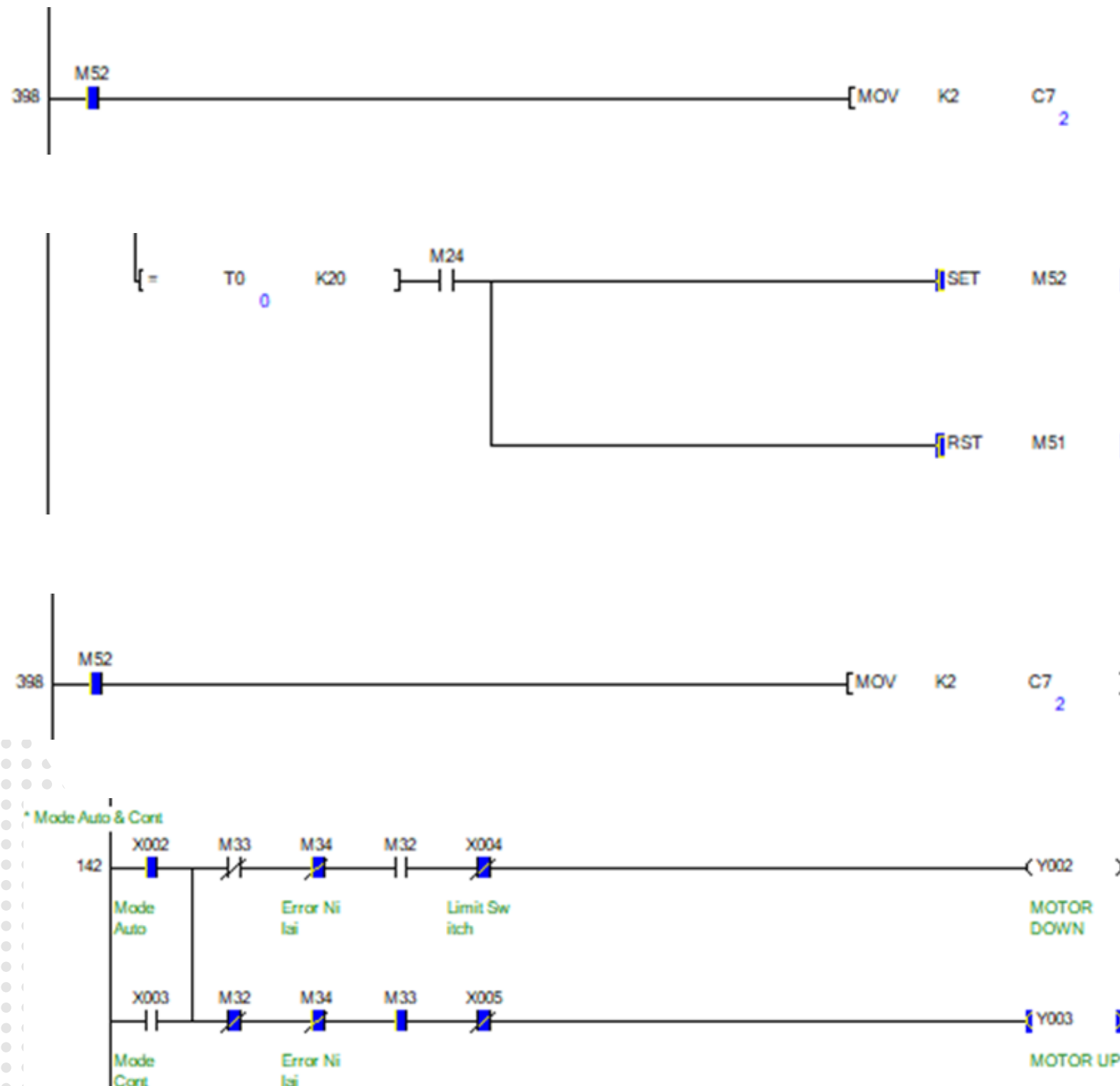
Simulasi Nilai ADC



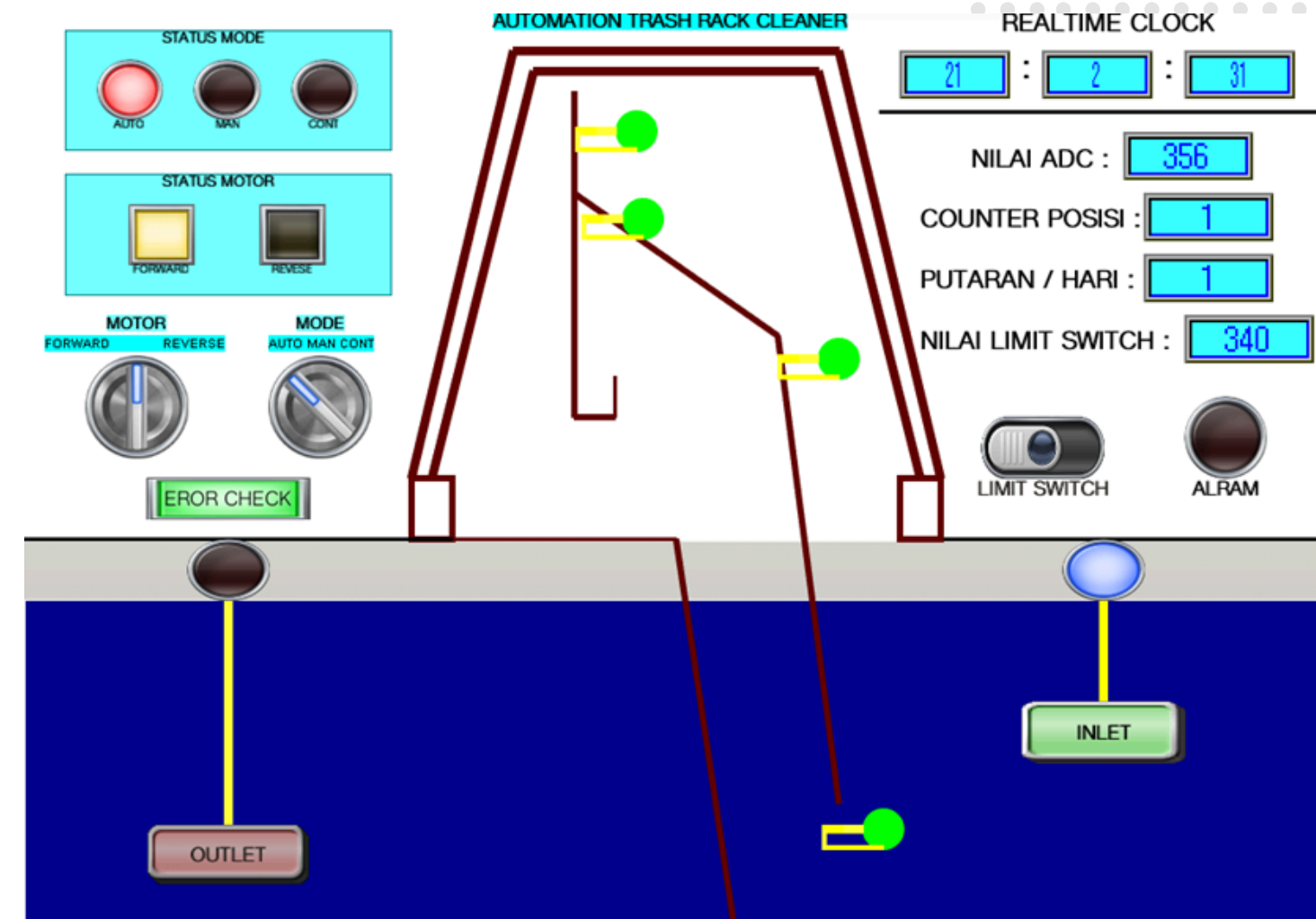
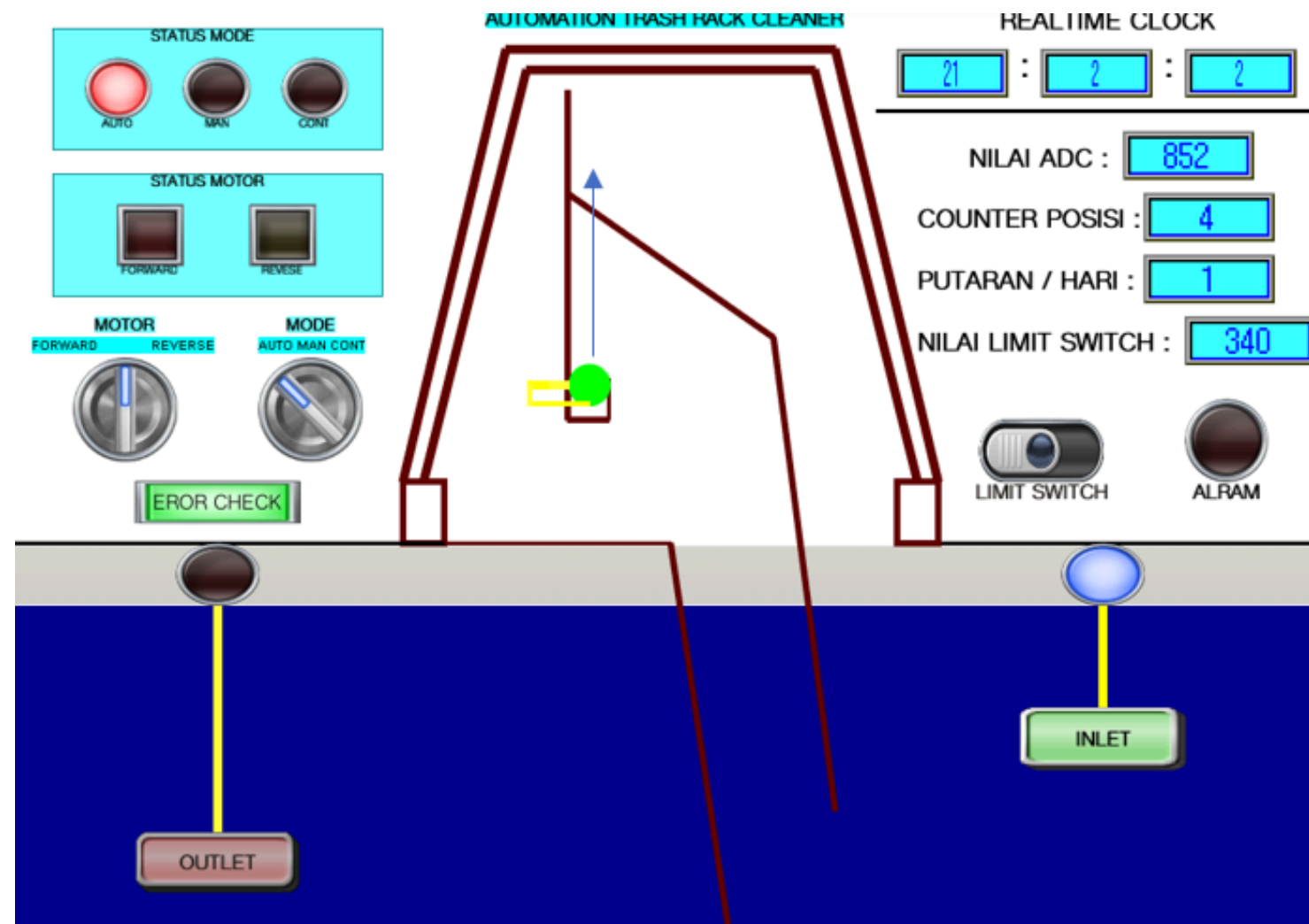
Simulasi Mode auto & Pergerakan penggaruk Turun



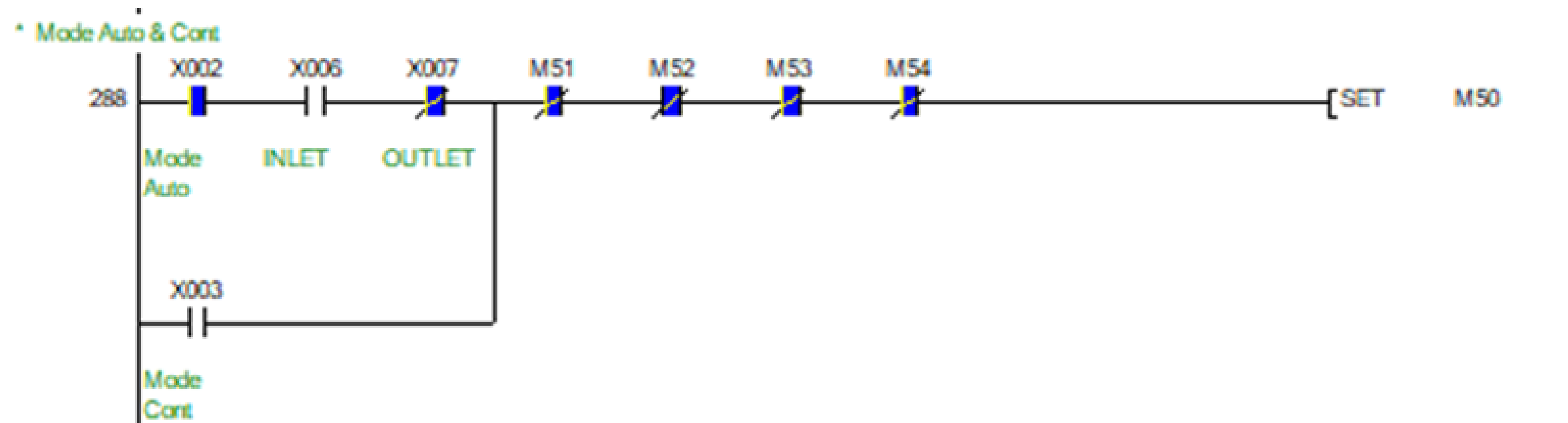
Pergerakan Penggaruk Naik



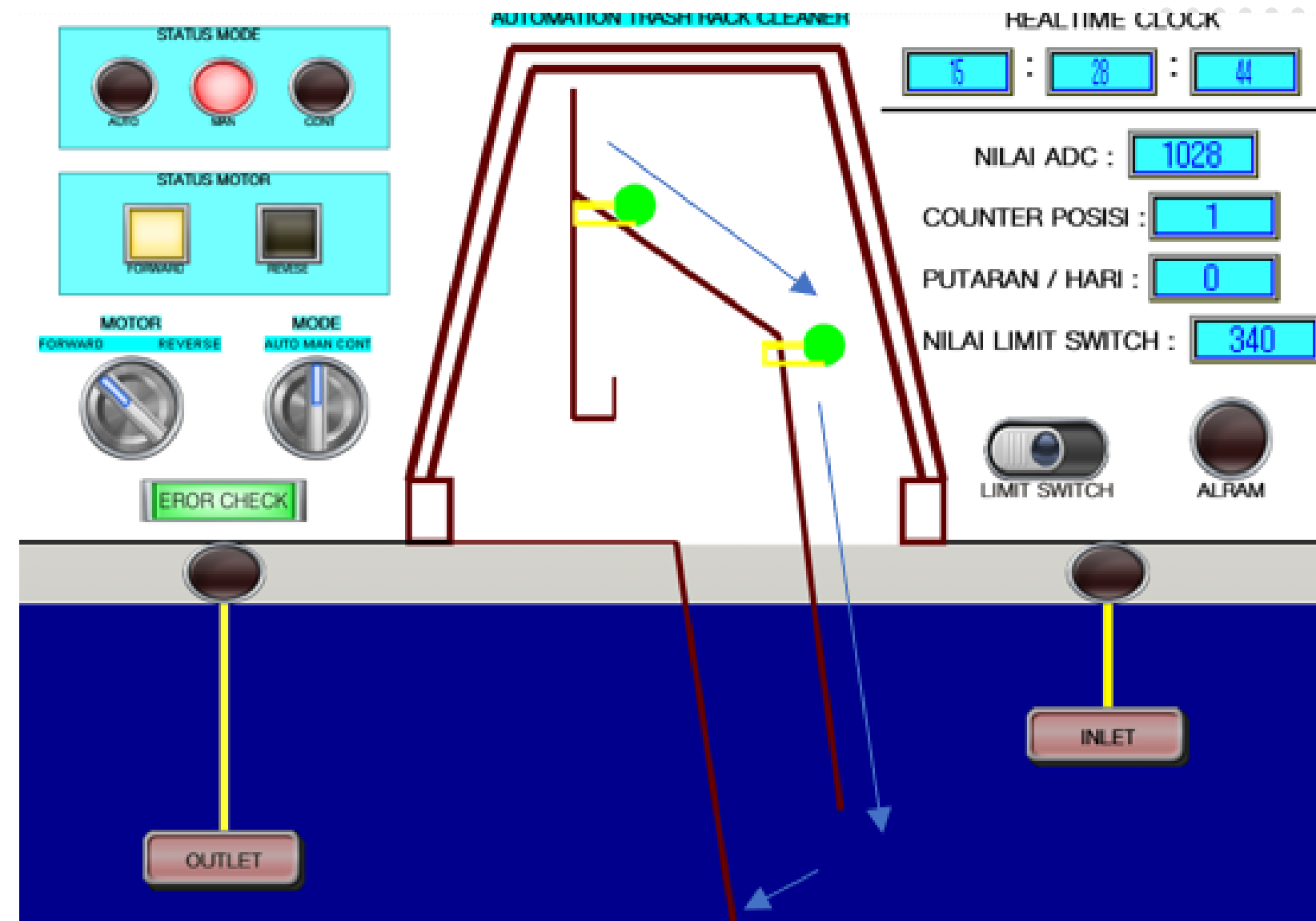
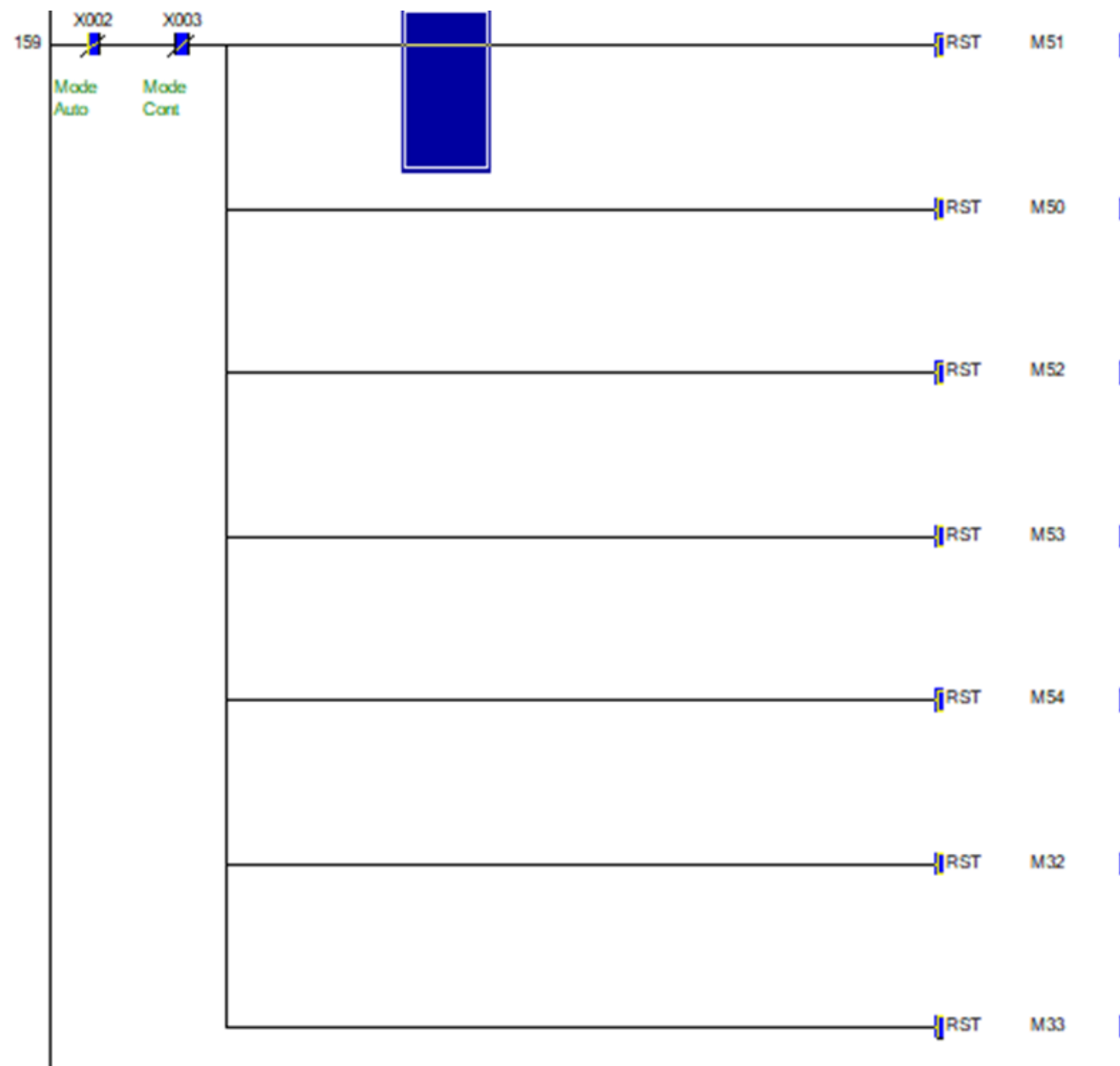
Pergerakan Penggaruk dari posisi Buang sampah ke posisi Standby



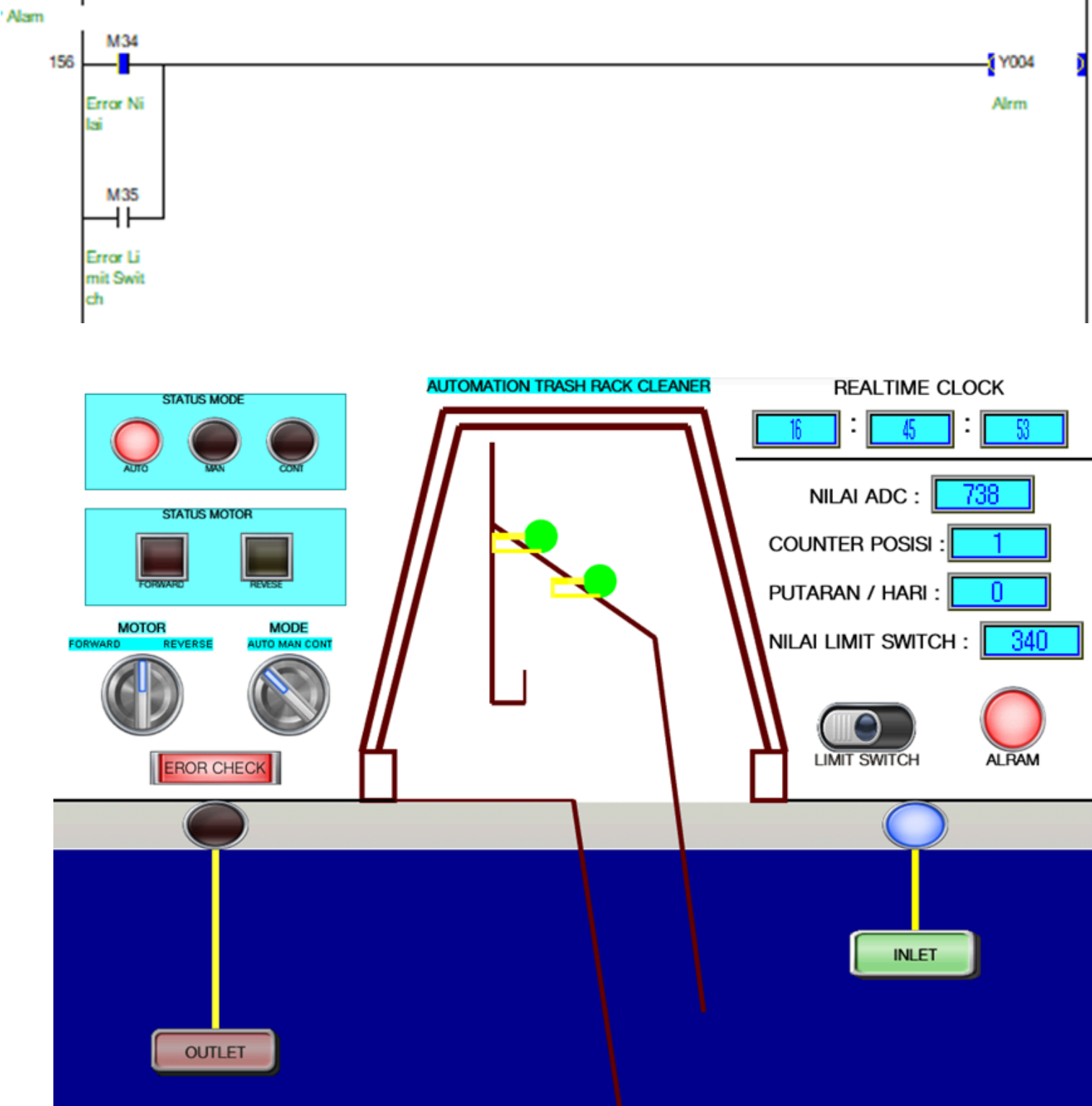
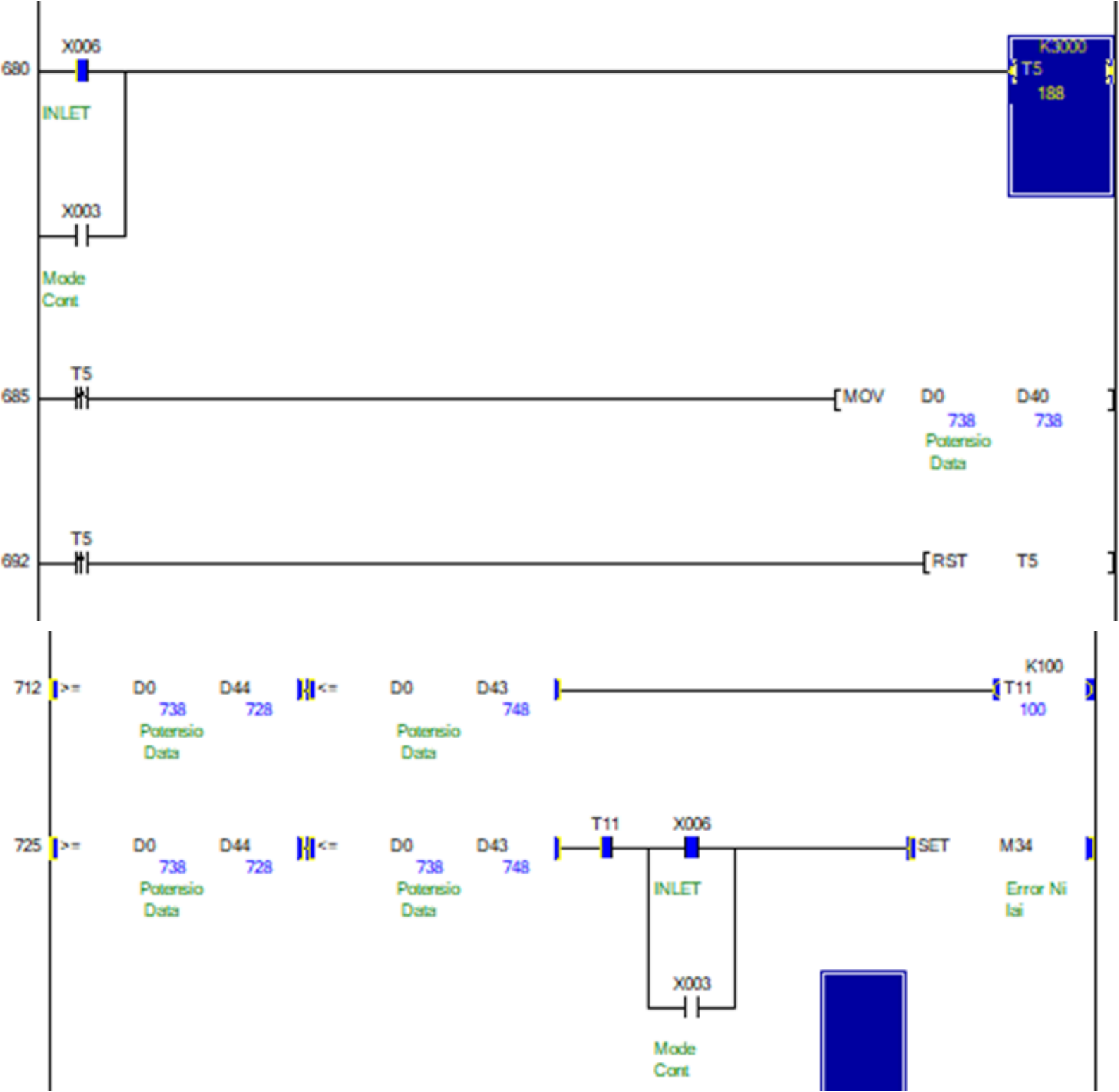
Mode Continue



Mode Manual



Error Check



Kesimpulan

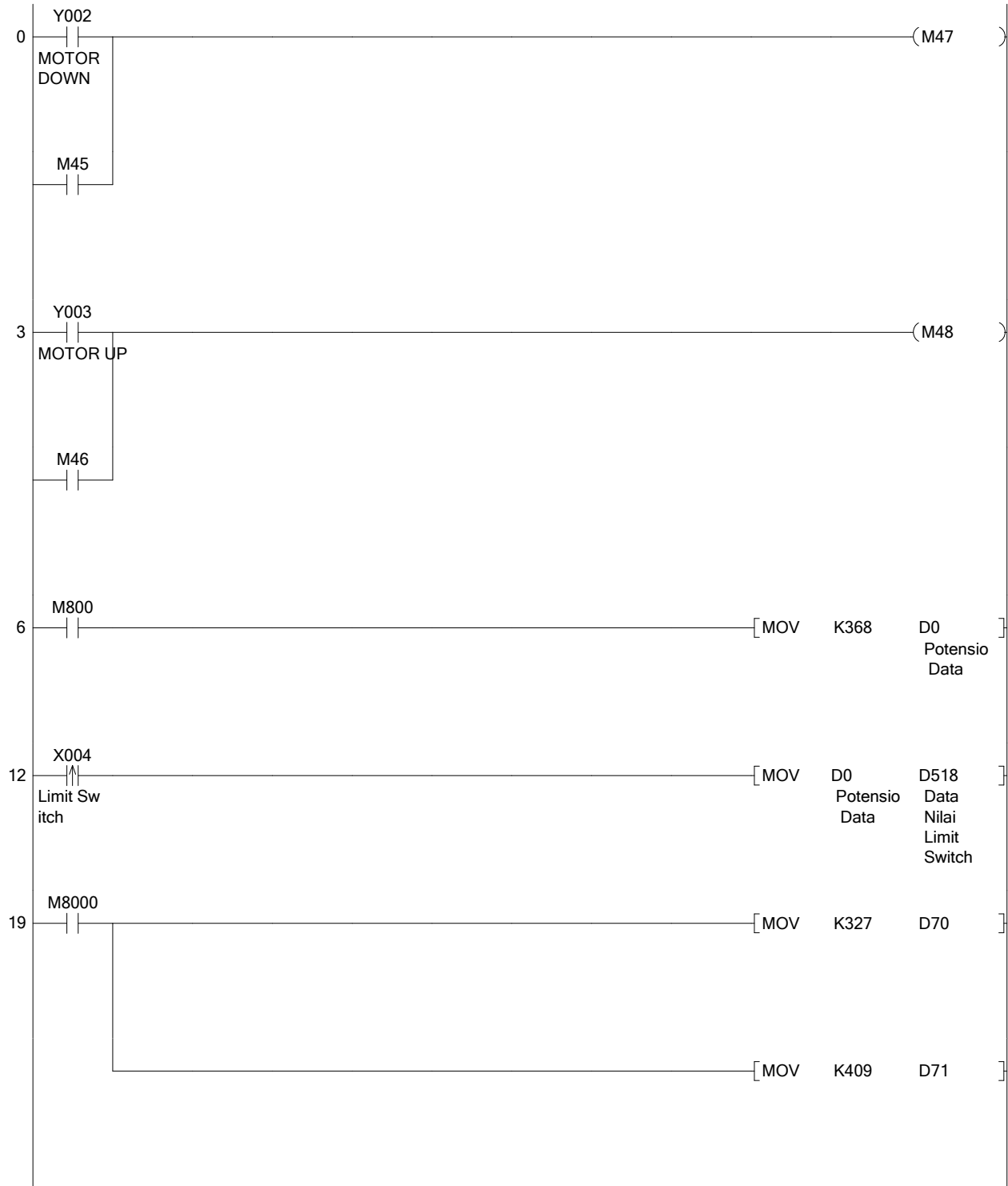
Simulasi Automation Trashrack Cleaner menggunakan GX Works dan GT Designer 3 berhasil menunjukkan pengoperasian sistem secara virtual. Pergerakan penggaruk, pengecekan kesalahan, dan pembacaan sensor berjalan sesuai logika. HMI menampilkan indikator posisi, mode operasi (Manual, Auto, Continuous), Sistem berfungsi sesuai perancangan dan merepresentasikan cara kerja Automation Trashrack



Terima Kasih

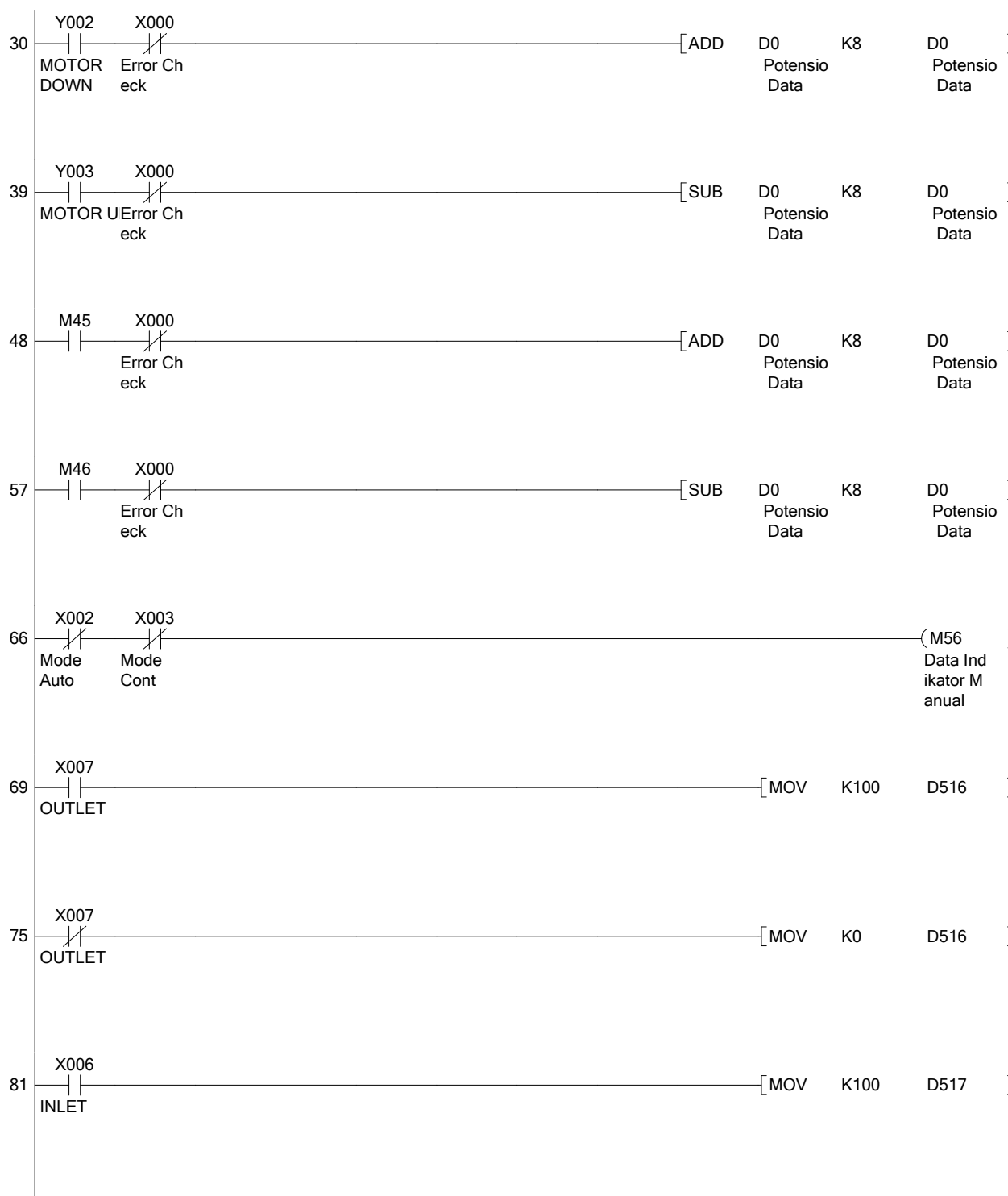
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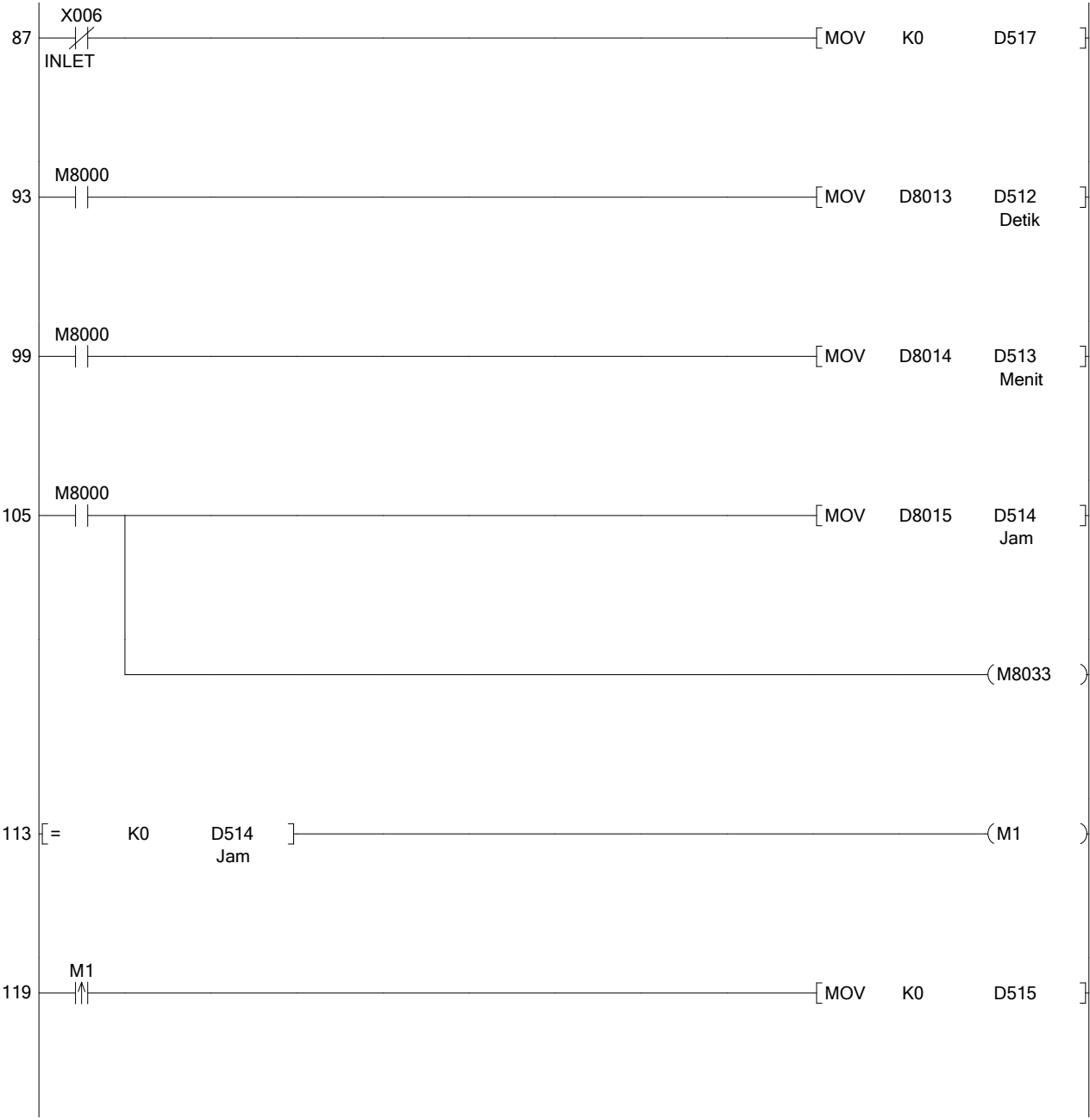
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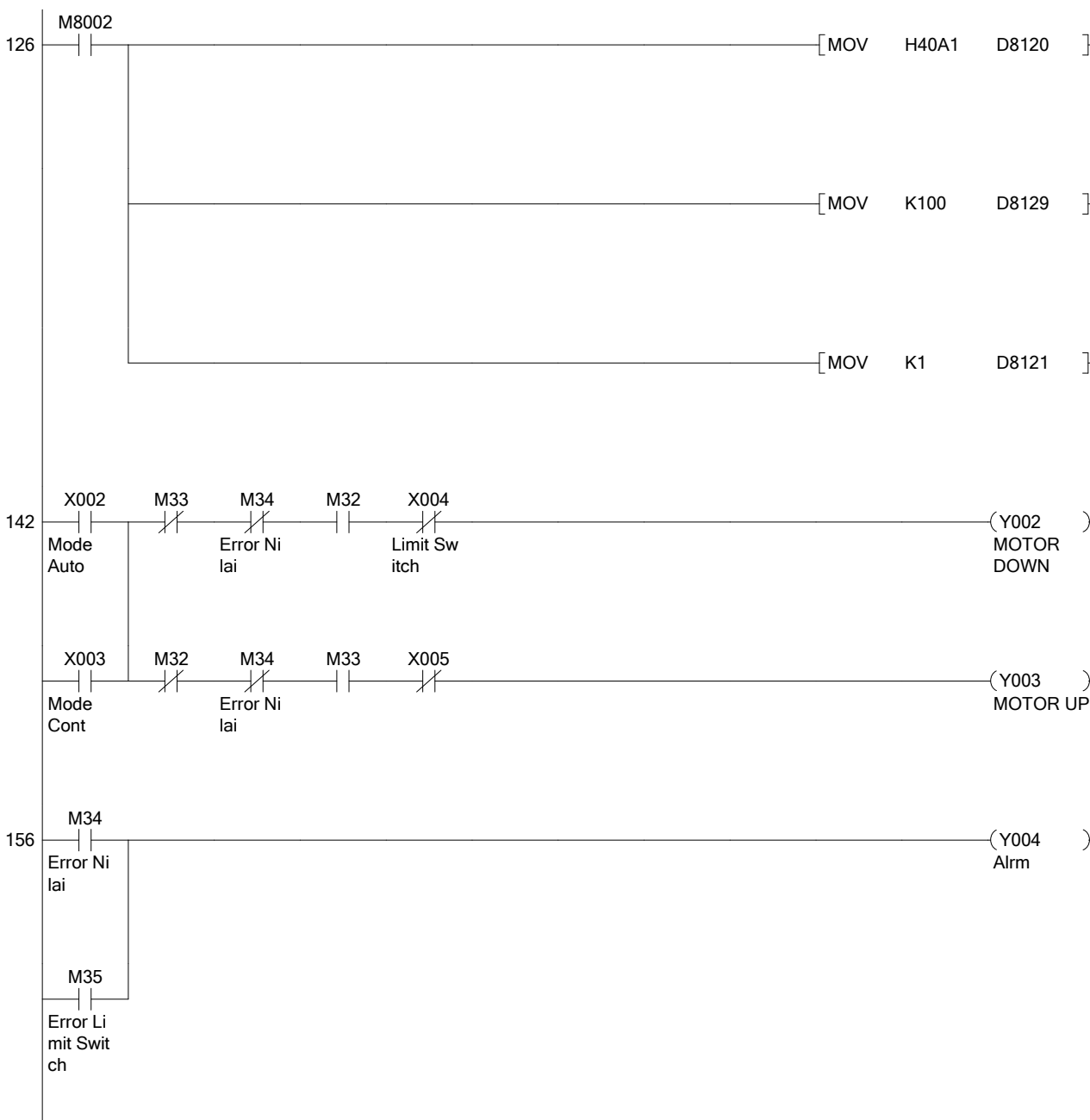
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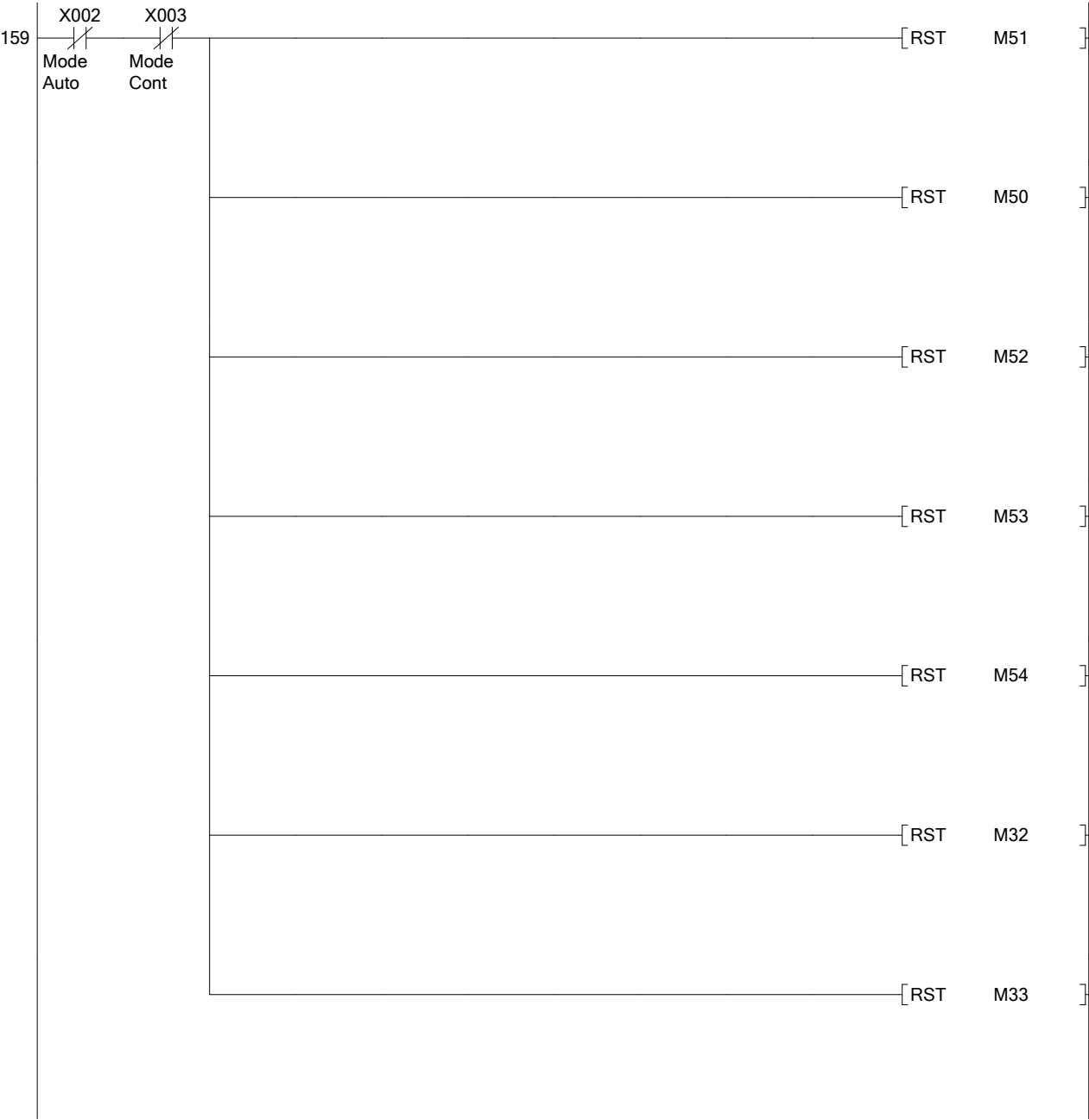
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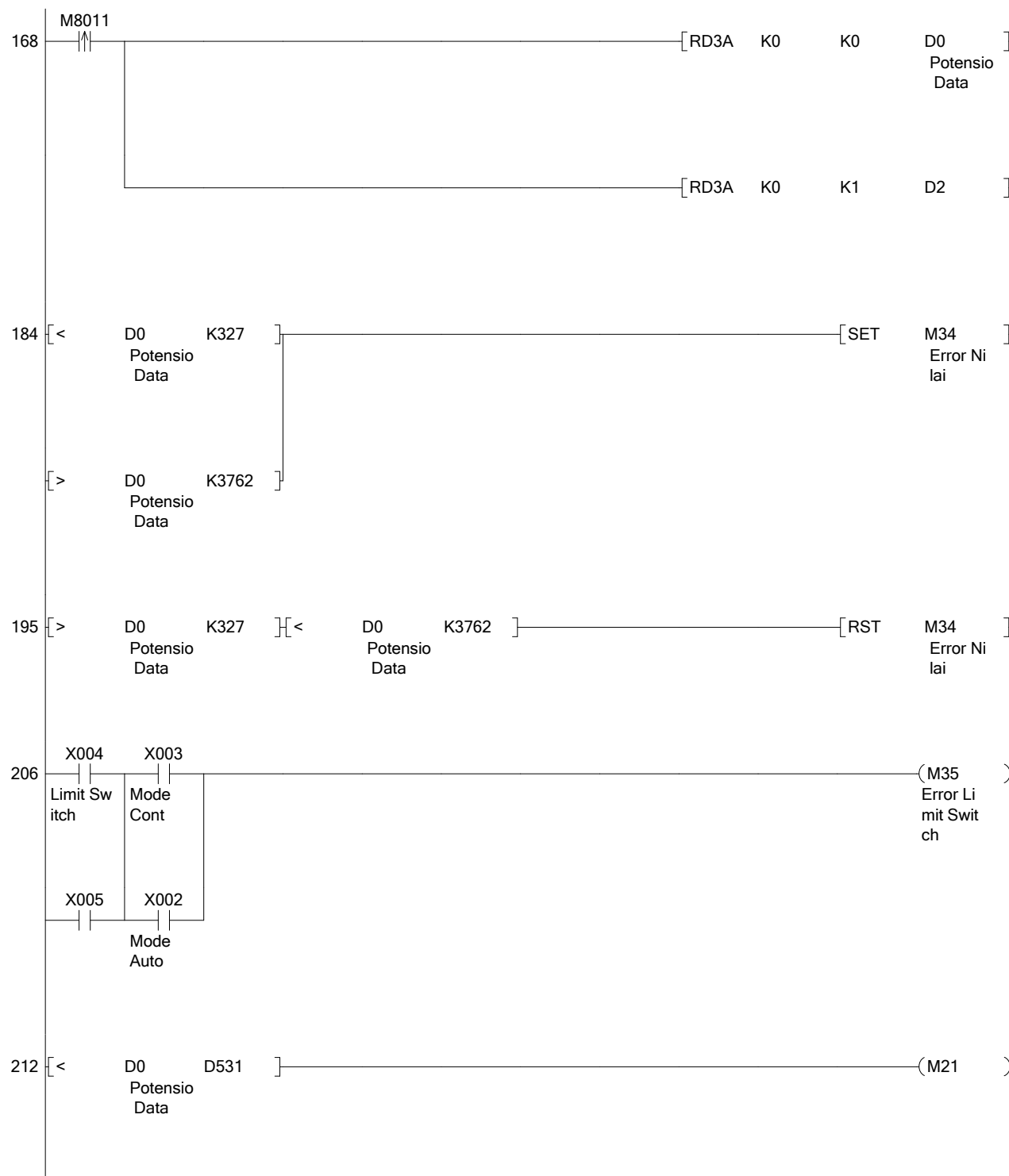
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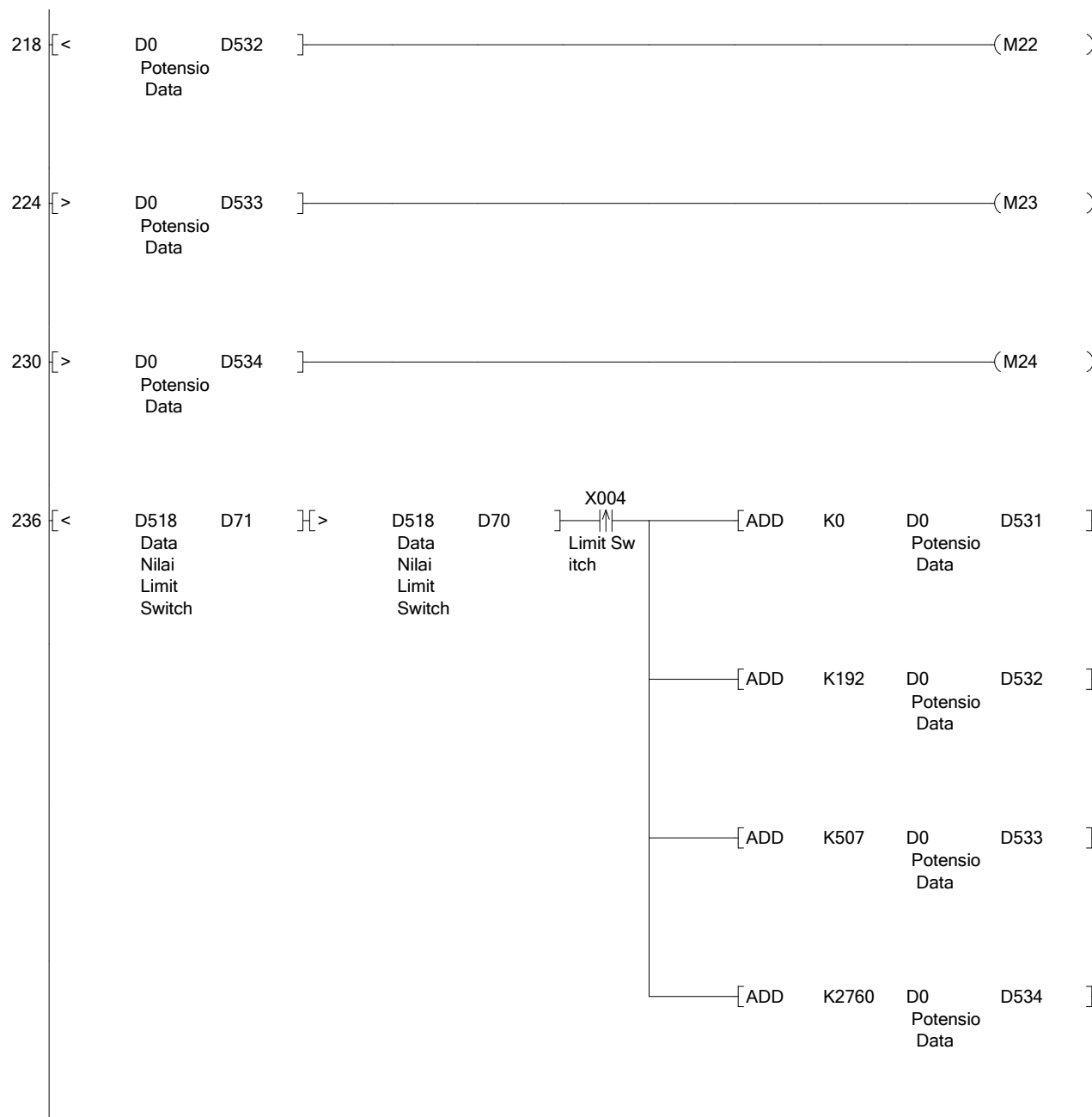
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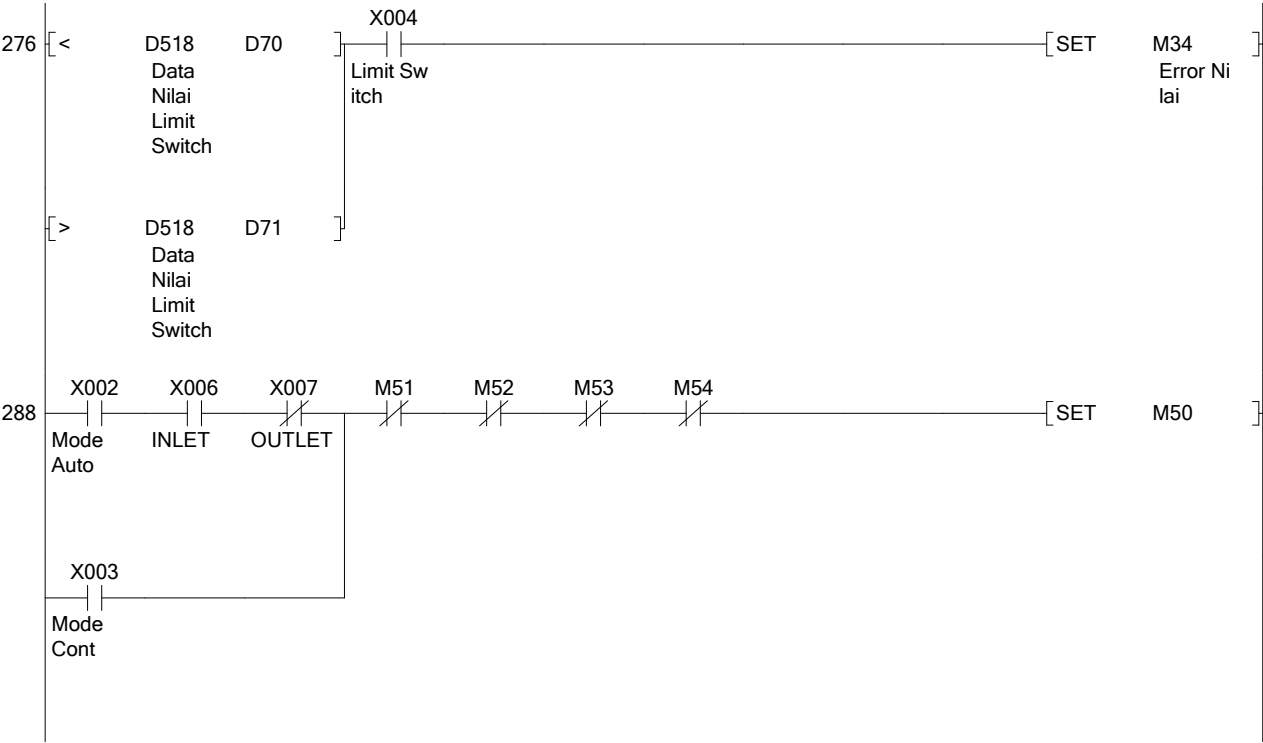
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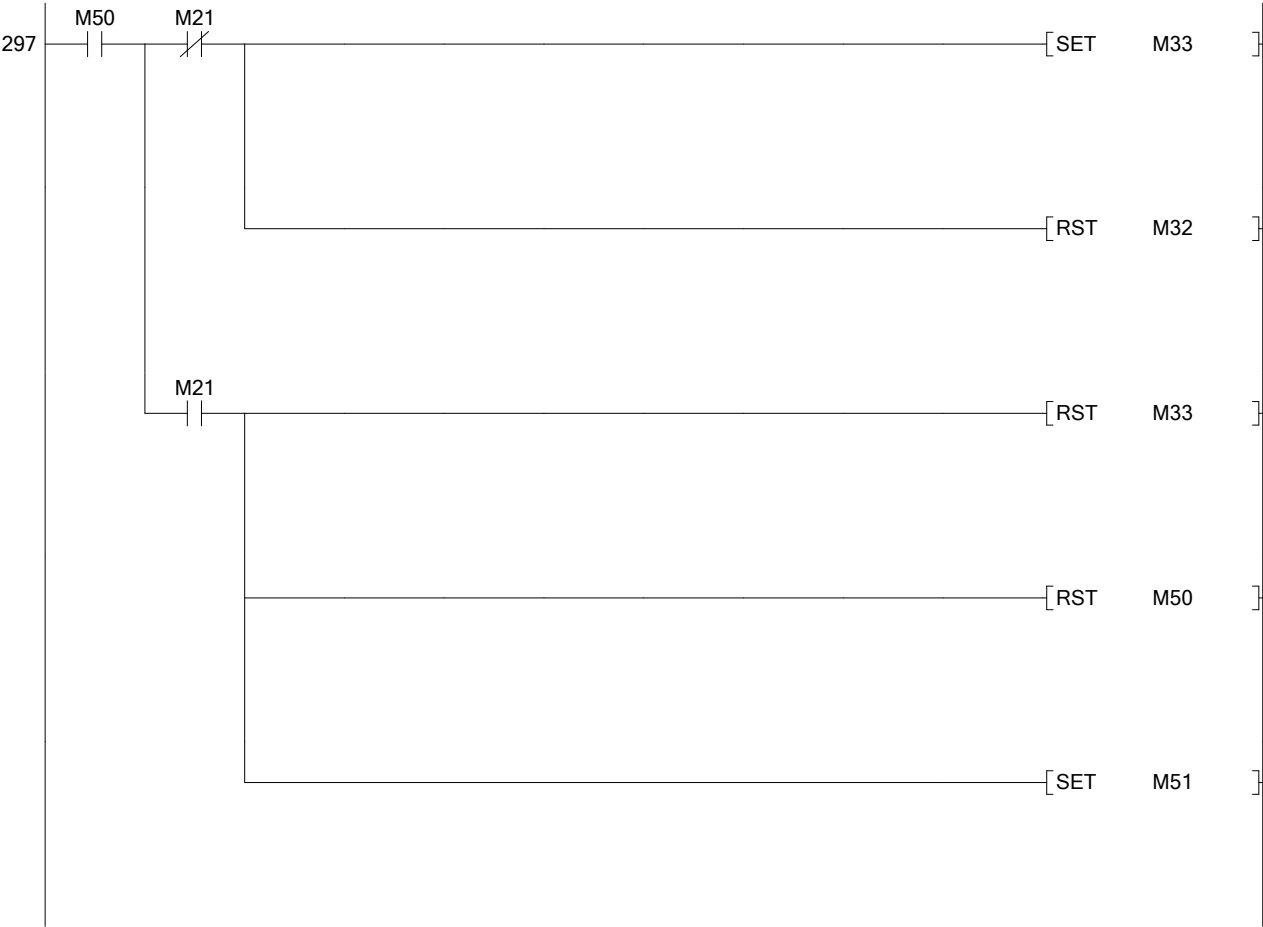


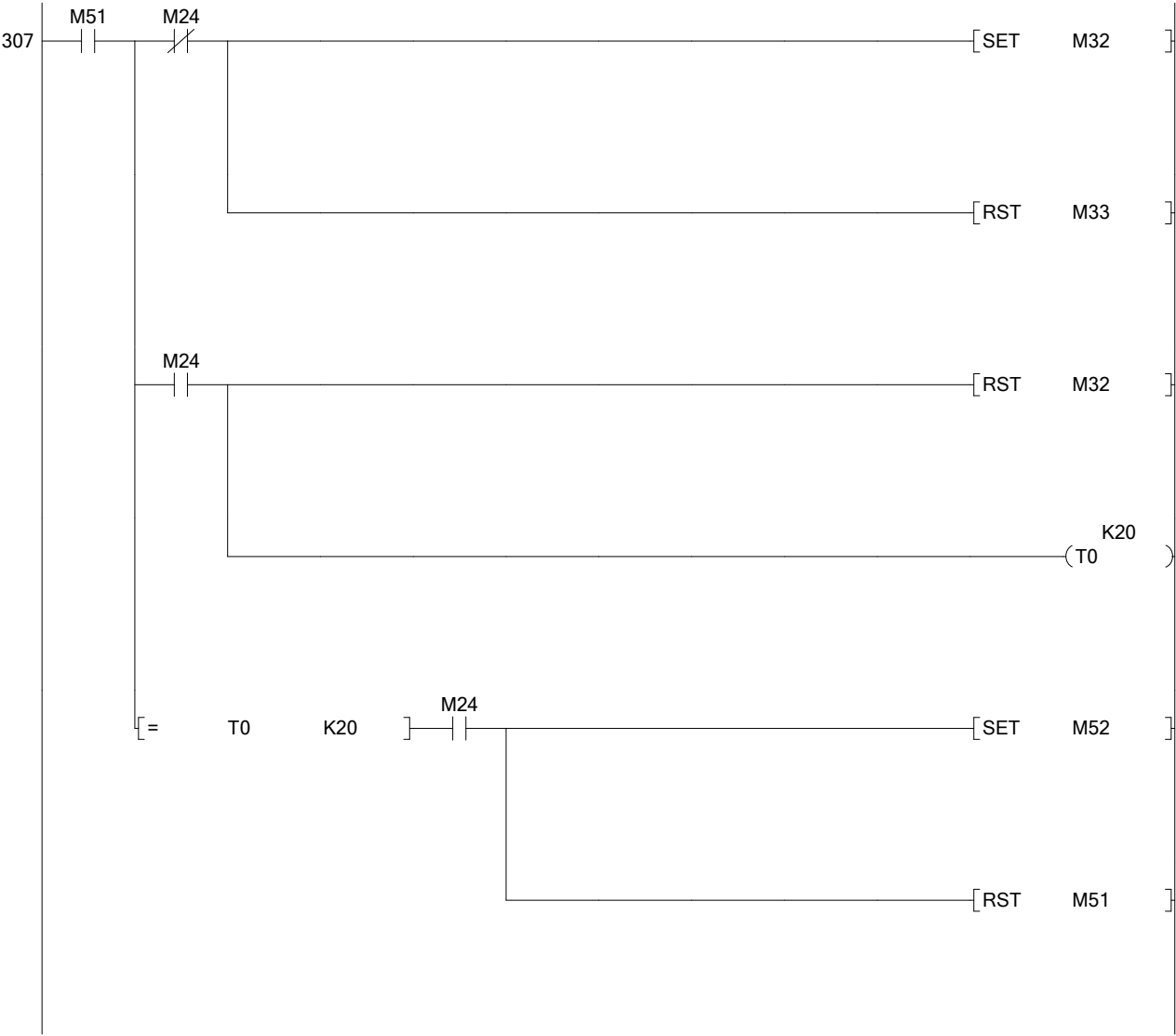
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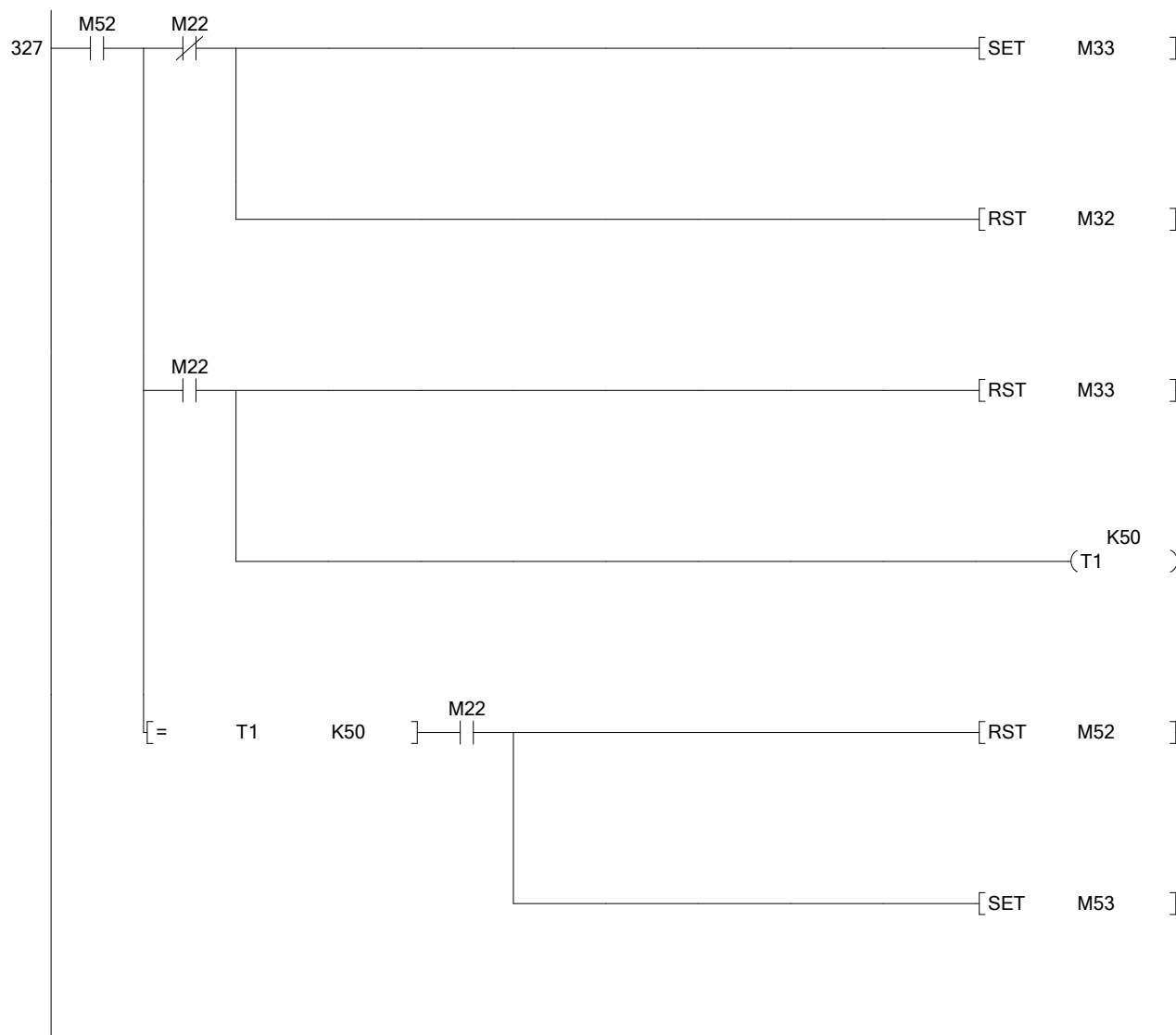


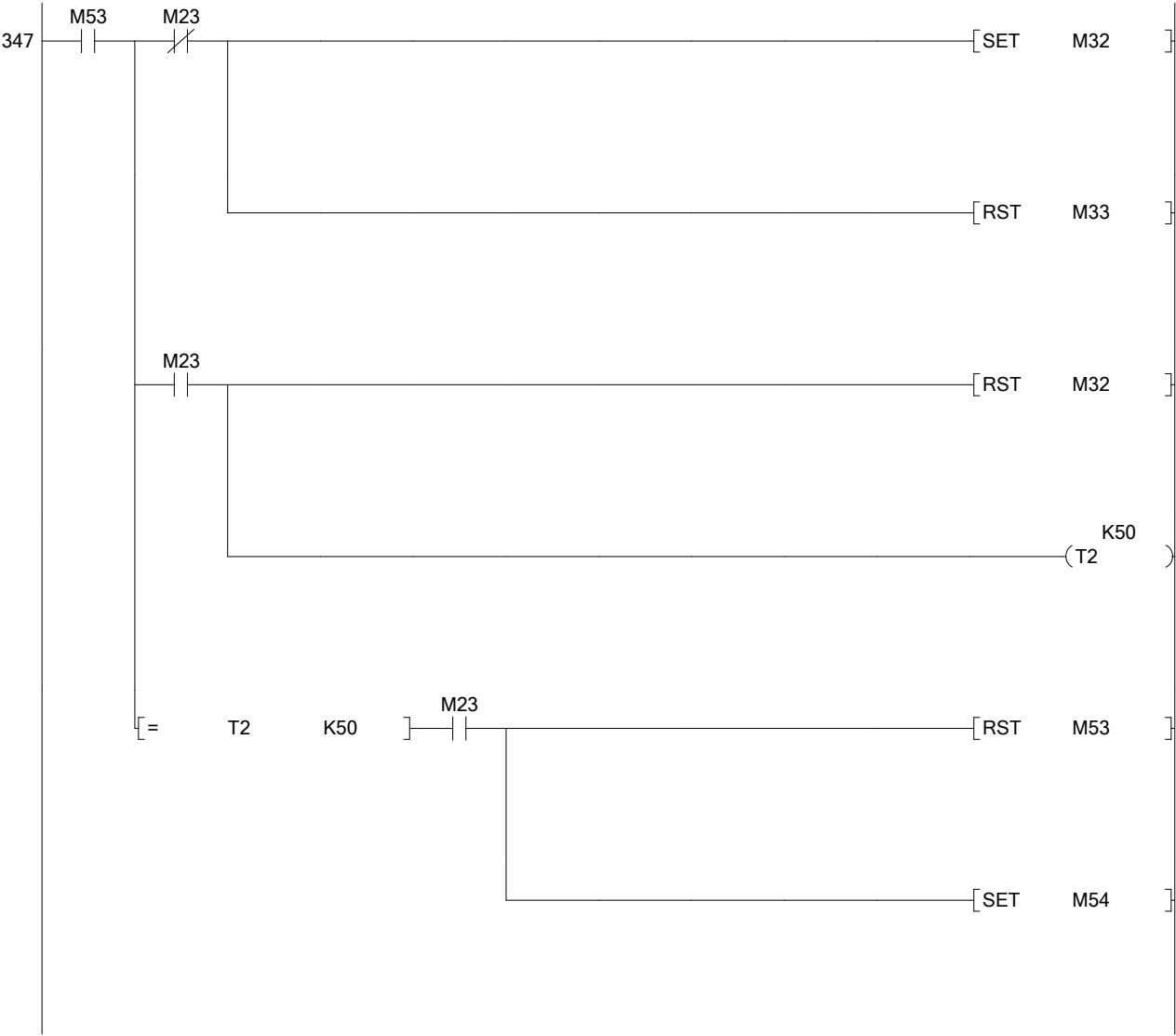


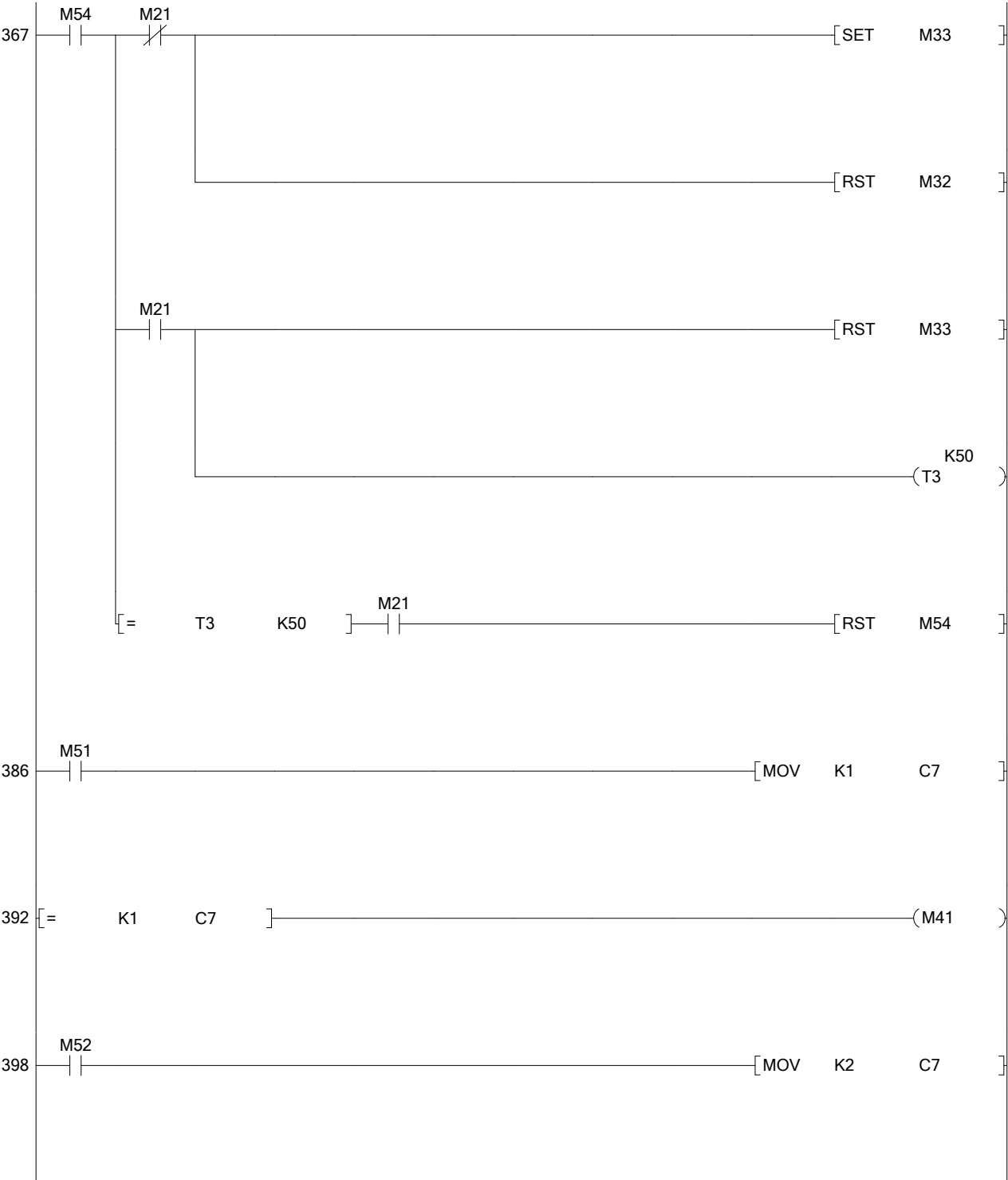


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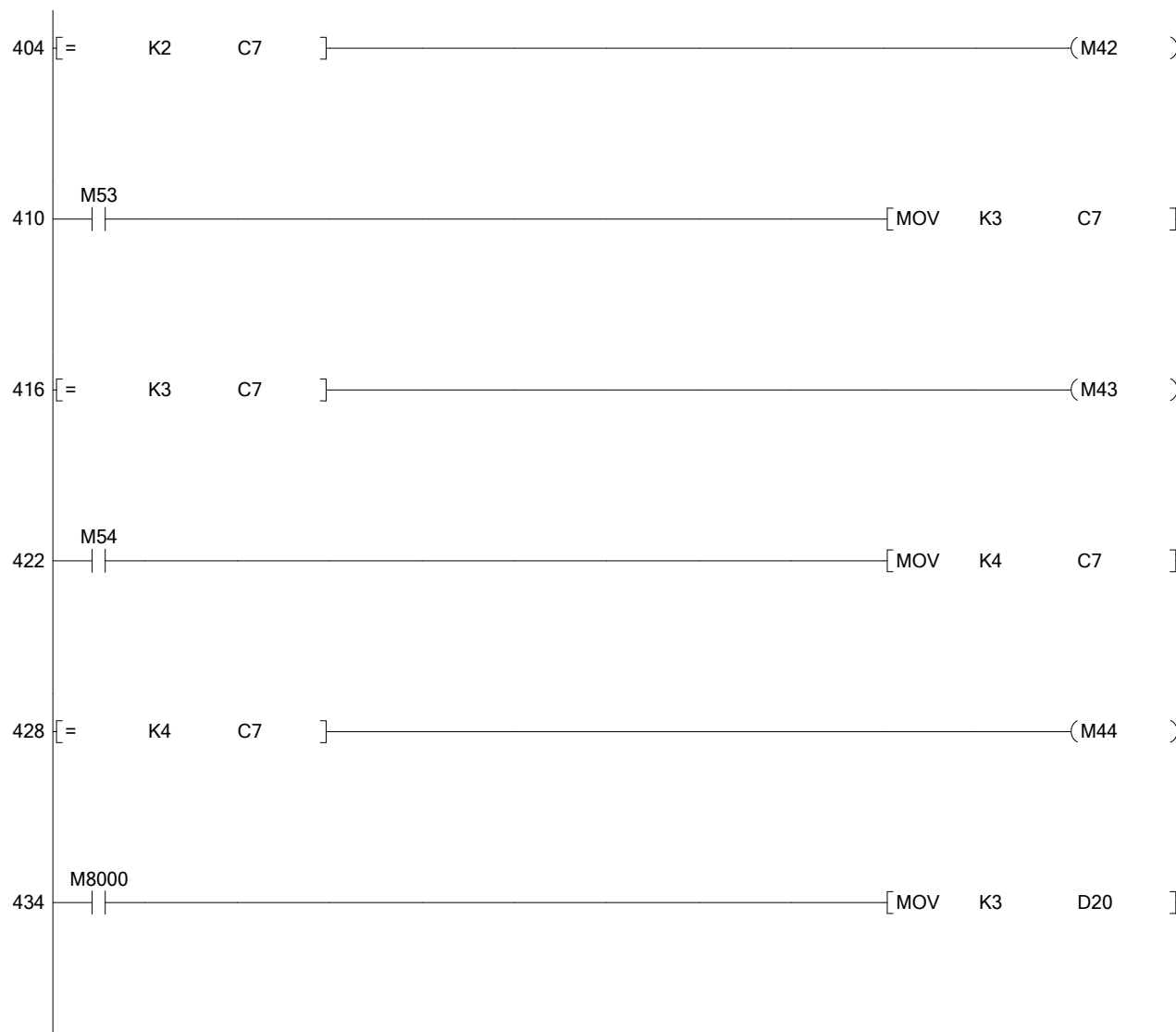






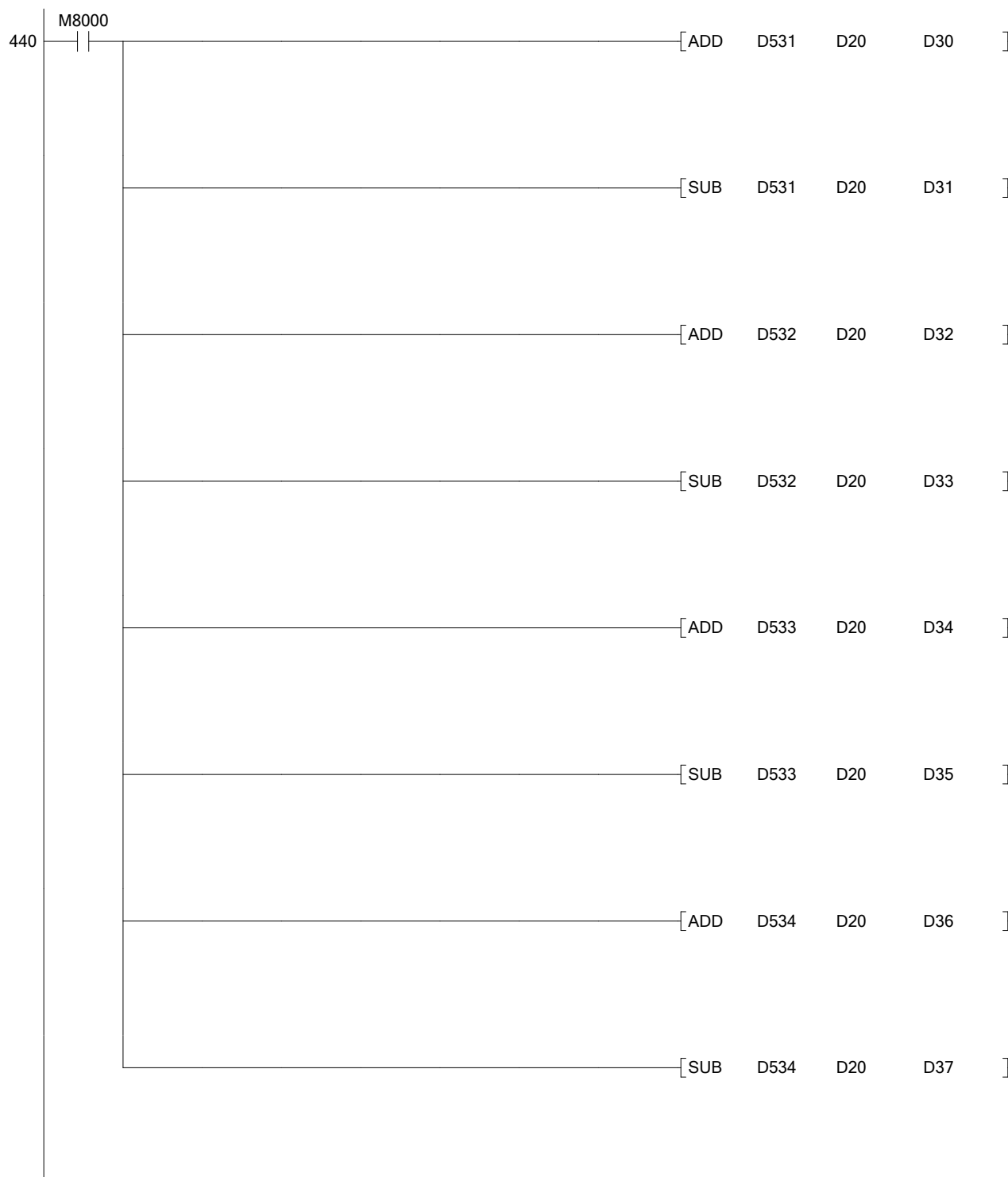
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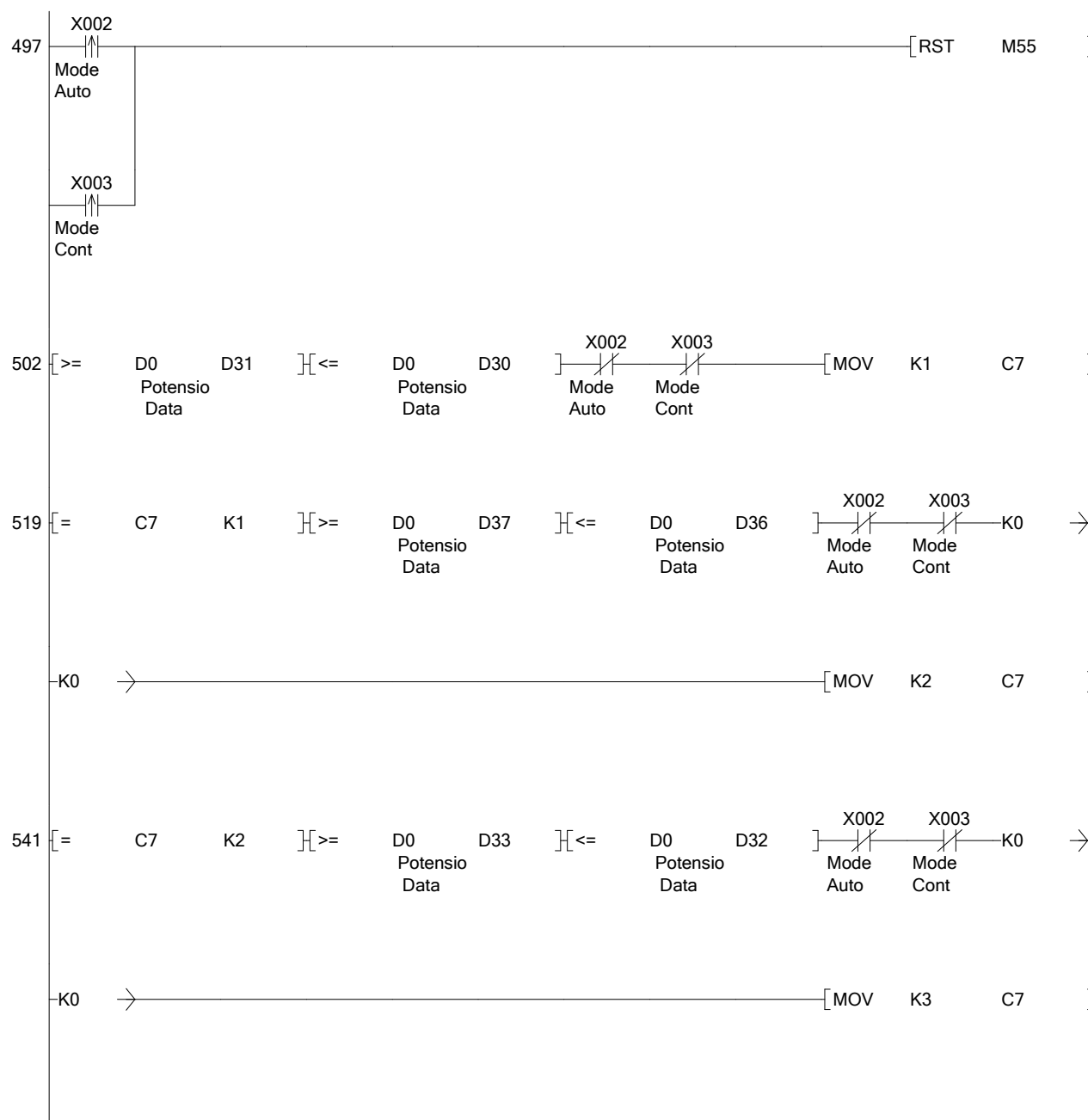
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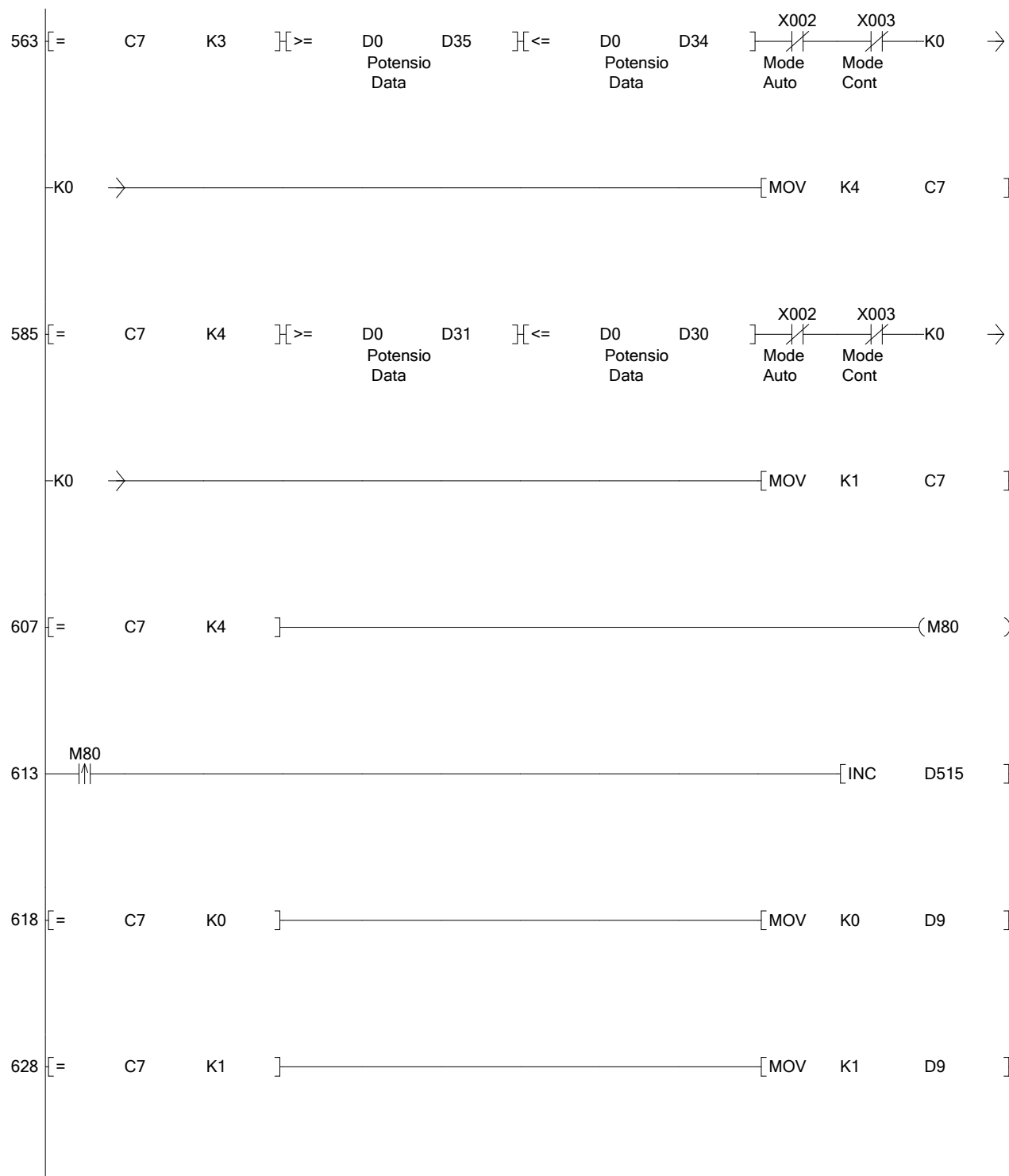
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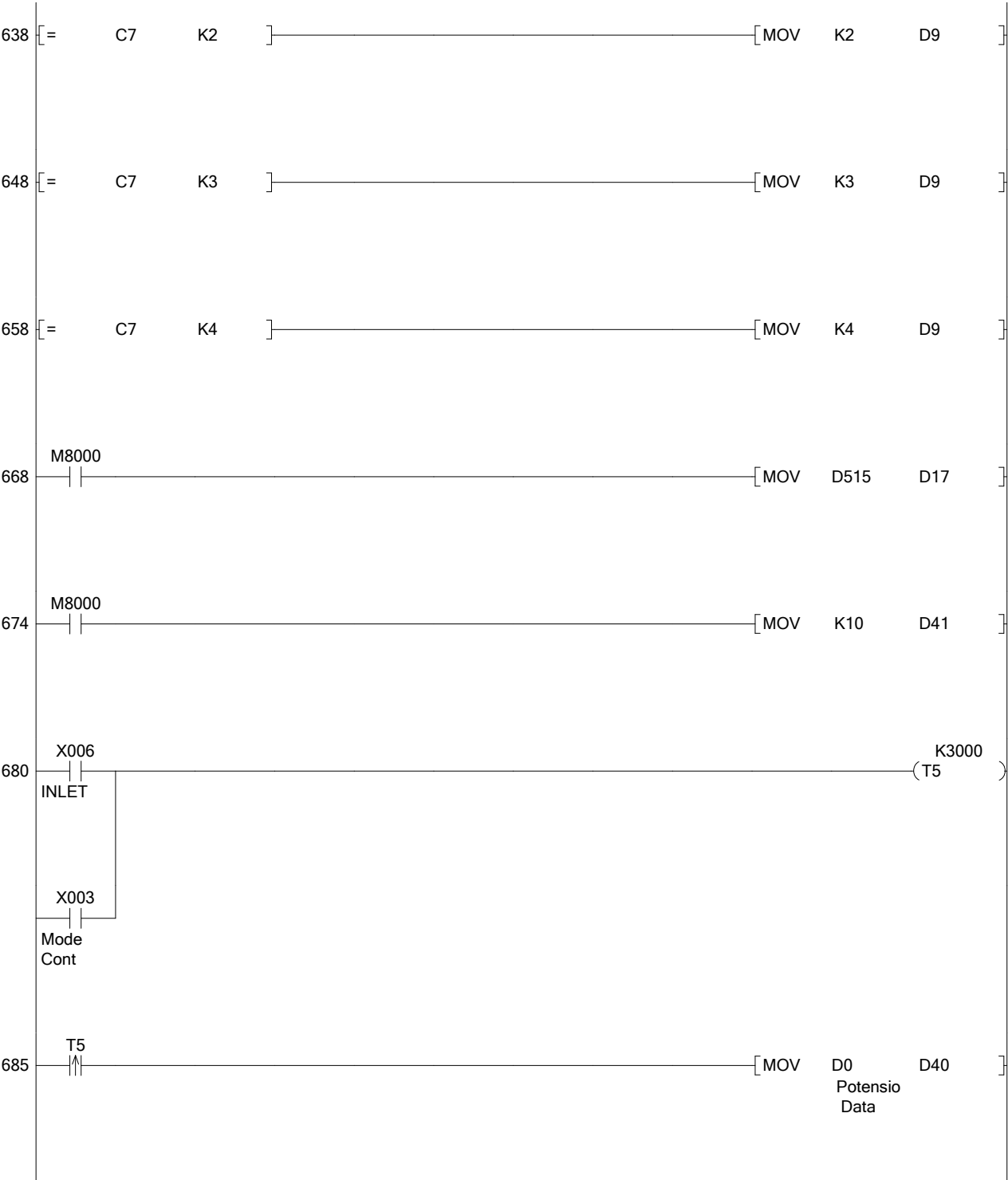
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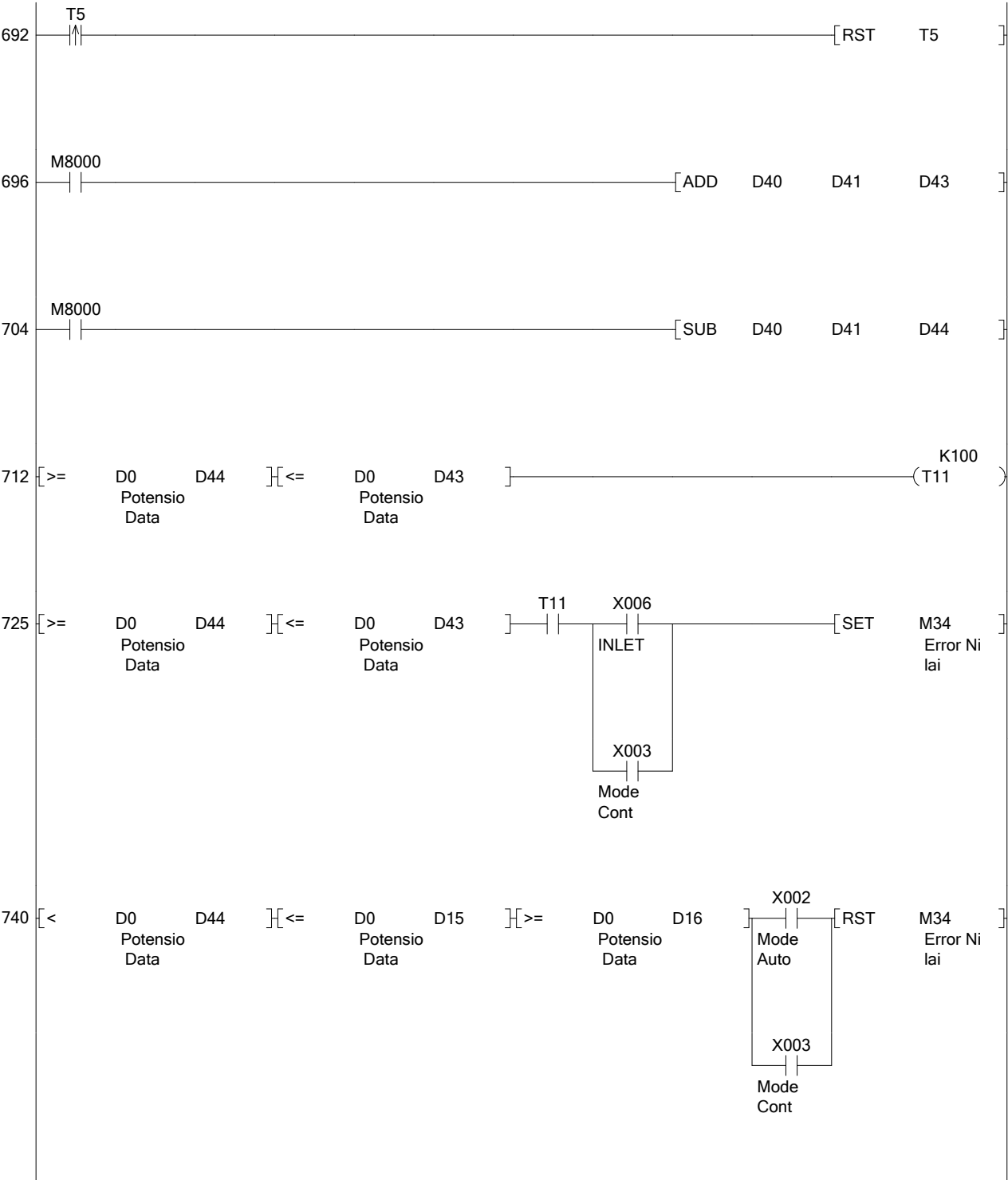


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